

The addicted predator–prey model: How substance abuse shapes productivity and growth-cycle dynamics*

Nara Chung[†] · Marwil J. Dávila-Fernández[†]

[†]*Colorado State University*

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Abstract

This paper extends Goodwin’s (1967) predator-prey growth-cycle model to incorporate the negative impact of substance abuse on labor productivity, establishing a novel link between health behavior and macroeconomic dynamics. Relying on a discrete-choice framework, workers are assumed to choose between drug-abusing and clean states in a mechanism intermediated by income distribution. A declining wage share raises the probability of substance abuse, while higher labor income reduces it. This behavioral health channel feeds back into the economy through a productivity damage function that amplifies or dampens cyclical fluctuations. By applying the existence part of the Hopf bifurcation theorem, we show that the resulting 3-dimensional nonlinear dynamic system admits endogenous, persistent, bounded fluctuations and the coexistence of limit cycles. Extending the framework to incorporate Minsky’s financial instability hypothesis allows us to further address how this process interacts with debt conditions. Taken together, our findings highlight how behavioral health deterioration can fundamentally reshape growth and real-financial market interactions in modern economies.

Keywords: Growth-cycles; Substance abuse; Endogenous Productivity; Minsky dynamics; Nonlinear systems; Financial instability; Heterogeneous agents

JEL: E12; E32; I14; I18; C61.

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1 Introduction

The sustained rise in drug-induced mortality across the United States since the late 1990s has revealed the macroeconomic dimensions of a health crisis long treated as a matter of individual behavior and public health policy. By the late 2010s, drug overdose deaths exceeded 70,000 annually, with pharmaceutical and synthetic opioids accounting for the majority of fatalities. The economic cost of opioid abuse alone has been estimated at well above \$500 billion annually when productivity losses, healthcare expenditures, criminal justice costs, and the value of lost life are aggregated (Florence et al., 2016; Ghimire et al., 2026). Beyond these static cost estimates, the epidemic has reshaped the composition and productivity of the U.S. workforce in ways likely to carry persistent macroeconomic consequences.

Yet standard macroeconomic models offer limited guidance on how the interaction between income distribution and health behavior feeds back into aggregate growth dynamics. In the Goodwin (1967) predator-prey growth-cycle model, the employment rate and the wage share oscillate endogenously due to the tension between capital accumulation and workers' bargaining power. This elegant two-dimensional framework captures the essential instability of capitalist economies in the tradition of Marx and Keynes, but it treats labor productivity growth as exogenous and the composition of the workforce as homogeneous. Neither assumption is tenable in an environment where income compression raises the probability of substance use disorder (OUD) and where OUD, in turn, depresses the effective productivity growth of the workforce.

This paper introduces substance abuse into the Goodwin framework as an endogenous channel linking income distribution to aggregate productivity dynamics. Relying on a discrete-choice framework in the spirit of Franke and Westerhoff (2017), workers are assumed to transition between drug-abusing and drug-clean states in response to the wage share. A declining wage share raises the probability of OUD, while a rising wage share reduces it. This behavioral health choice feeds into the economy through a productivity damage function $D(z)$, where z is a population composition index measuring the prevalence of OUD among workers. The key implication is that the standard Goodwin growth cycle is amplified by the health channel: in the low-wage-share phase, both capital accumulation and rising OUD prevalence simultaneously push employment upward; in the high-wage-share phase, both forces jointly dampen employment. This dual amplification mechanism breaks the conservative structure of the original Goodwin model and, above a critical bifurcation threshold, gives rise to self-sustaining limit cycles.

Before developing the theoretical framework, we present a brief empirical examination of U.S. state-level data covering 1998–2019 to ground the model's key mechanisms in the data. Annual labor share changes are classified into nine severity bins—from very large negative shocks through no-shock years to very large positive shocks—and their relationship with age-adjusted drug-induced mortality is traced using event-style local projections following Teulings and Zubanov (2014). The static dose-response pattern confirms a broadly monotone relationship: states experiencing severe negative labor share shocks exhibit drug mortality substantially above the sample mean, while states with positive shocks show mortality below it. The dynamic local projections further indicate that moderate positive labor share gains are associated with statistically meaningful reductions in drug-induced deaths over horizons of one to four years, consistent with the income-protective channel formalized in the model.

On the theoretical side, we establish three main results. First, the three-dimensional dynamic system comprising the employment rate, the wage share, and the workforce composition index admits a unique economically meaningful interior equilibrium under mild regularity conditions (Propositions 1 and 3). Second, applying the existence part of the Hopf bifurcation theorem, we show that as the autonomous wage-growth parameter β_0 crosses a

critical threshold, the equilibrium loses local asymptotic stability and the system admits a family of periodic solutions (Propositions 2 and 4). Third, numerical simulations above the bifurcation threshold reveal the coexistence of multiple stable limit cycles whose amplitude depends on initial conditions rather than structural parameters alone. This history dependence implies that economies experiencing larger initial macroeconomic imbalances may lock into high-amplitude growth cycles with wider oscillations in employment, the wage share, and workforce health composition.

The model further generates sharp comparative dynamics with respect to two key parameters. The intensity of choice parameter γ , governing the sensitivity of substance-use behavior to income fluctuations, acts as an amplifier: a higher γ widens the limit cycle orbits, while a lower γ —achievable through strong social safety nets, universal healthcare access, or mental health services—compresses them. By contrast, the productivity damage parameter θ generates a counterintuitive stabilizing effect: higher θ strengthens a self-correcting feedback loop through which OUD-induced productivity losses trigger faster employment and wage-share recovery, ultimately compressing the cycle amplitude and eliminating the coexistence of multiple attractors.

The remainder of the paper is organized as follows. Section 2 reviews the empirical literature on drug abuse, labor productivity, and labor force participation, before providing a preliminary look at the data. Section 3 develops the theoretical model. Section 4 presents numerical experiments. Section 5 offers concluding remarks. Mathematical proofs are collected in the Appendix.

2 An overview of the literature

2.1 Drug abuse and labor productivity

A broad empirical literature establishes that drug abuse significantly reduces labor productivity through multiple channels, including impaired physical and cognitive functioning, absenteeism, presenteeism, long-term disability, and premature mortality. These effects lower effective labor input not only by reducing output per worker but also by permanently removing embodied human capital from the economy.

Early U.S. evidence is provided by [Florence et al. \(2016\)](#), who estimate that prescription opioid overdose, abuse, and dependence imposed an economic burden of \$78.5 billion in 2013, with approximately \$20 billion attributable to lost productivity arising from reduced working hours, disability, incarceration, and premature death. Importantly, productivity losses include both market labor and household production, indicating that the economic damage of substance abuse extends beyond formal employment relationships.

More recent estimates by [Ghimire et al. \(2026\)](#) show that morbidity-related productivity losses attributable to substance use disorder (SUD) reached \$92.65 billion in the United States in 2023, with inability to work accounting for nearly half of total losses. In this paper, morbidity-related productivity losses refer to the annual value of production forgone because substance use disorder reduces individuals' ability to work while they remain alive, through absenteeism, presenteeism, and inability to work. These losses exclude premature mortality and instead capture recurrent, potentially reversible reductions in effective labor productivity. This makes morbidity a natural channel through which health deterioration affects growth dynamics, operating as an endogenous productivity wedge rather than a one-time level shock. Mortality-based estimates represent an upper bound on productivity losses, as premature death permanently eliminates all future labor contributions, whereas morbidity reduces productive capacity among the living. Using Canadian data, [Cheung et al.](#)

(2023) estimate that opioid overdose deaths between 2016 and 2019 generated \$8.8–\$10.9 billion in aggregate lost future labor productivity, corresponding to roughly \$0.6–\$0.7 million per death under a human capital approach, and that more than two-thirds of overdose victims were employed within five years prior to death. These findings challenge the notion that opioid-related harm is concentrated among marginal or unproductive workers.

These empirical findings map directly into the endogenous productivity channel of the extended Goodwin growth-cycle framework in the next section. In the Goodwin model with substance abuse, labor productivity grows at an exogenous technological rate but is discounted by a health damage function that depends positively on the prevalence of drug abuse among workers:

$$\frac{\dot{a}}{a} = [1 - D(z)]\alpha,$$

where z captures the composition of the labor force between drug-abusing and drug-clean workers, and $D'(z) > 0$. This formulation captures how increased substance abuse can reduce aggregate productivity growth even when employment remains unchanged.

Occupational evidence further reinforces this mechanism. Lundstrom et al. (2025) document that stimulant co-involvement in synthetic opioid overdose deaths is especially prevalent in physically demanding occupations such as construction, extraction, and agriculture. Qualitative evidence suggests that stimulant use is often adopted to temporarily sustain alertness and productivity despite opioid-induced impairment. Within the model, such behavior corresponds to a short-run masking of productivity loss, followed by stronger long-run damage through increases in z and a higher productivity discount $D(z)$, amplifying cyclical fluctuations.

Overall, the literature provides strong empirical support for modeling substance abuse as an endogenous drag on productivity growth that can reshape the amplitude and persistence of macroeconomic cycles.

2.2 Drug abuse and labor force participation rate

A closely related literature documents that drug abuse substantially reduces labor force participation, particularly among prime-age workers, by increasing disability, worsening health, and weakening labor market attachment. Participation losses therefore reflect both individual health constraints and broader macroeconomic feedback effects.

Krueger (2017) provides foundational evidence linking opioid use and pain to reduced labor force participation, showing that nearly half of prime-age men out of the U.S. labor force consume pain medication daily, most commonly prescription opioids. Labor force participation declined more sharply in areas with higher opioid prescribing rates, consistent with health-driven labor market exit rather than prolonged unemployment spells.

Quantifying these effects during the COVID-19 period, Greenwood et al. (2022) estimate that increased substance abuse accounts for between 9 and 26 percent of the decline in prime-age labor force participation between February 2020 and June 2021. Individuals with opioid or methamphetamine use disorders exhibit labor force participation rates 13–16 percentage points lower than those of non-abusers.

Mortality represents the most extreme form of participation loss. Cheung et al. (2023) show that opioid overdose deaths in Canada disproportionately affected working-age individuals who were strongly attached to the labor market prior to death, implying permanent and non-reversible participation losses. Occupational patterns documented by Lundstrom et al. (2025) further indicate that overdose deaths are concentrated in labor-intensive sectors, intensifying participation declines in industries with high baseline employment attachment.

Empirical evidence further suggests that the relationship between labor market conditions and substance abuse depends not only on aggregate employment or wage growth, but on their distributional and sectoral composition. Using a Bartik-style identification strategy, ? show that employment and wage growth in lower- and middle-wage industries—those more likely to employ less-educated workers—are associated with reductions in opioid overdose mortality, whereas employment growth in higher-wage industries can be associated with increases in overdose deaths. These findings indicate that economic growth is not uniformly protective and that shifts in income distribution and job structure can alter substance use risk across populations. As such, labor market dynamics influence substance abuse through relative income and opportunity channels, rather than through aggregate conditions alone.

These empirical findings correspond closely to the behavioral and compositional channel in the Goodwin model with substance abuse. Rather than treating labor force participation as exogenous, the model introduces a population composition index z that evolves endogenously with income distribution. Declines in the wage share increase the probability of substance abuse, while higher labor income reduces it, consistent with empirical evidence linking economic stress to substance use.

Through this mechanism, participation losses emerge endogenously: increases in z reduce productivity growth, which dampens employment expansion and reinforces labor market exit. Participation declines thus arise not only through direct disability or withdrawal from the labor force but also through macroeconomic feedback loops operating via productivity and employment dynamics.

Causal evidence from Beheshti (2023), who shows that exogenous reductions in opioid availability increased labor force participation through employment gains, further supports this structure. In the model, such policies reduce z , weaken the productivity damage function $D(z)$, and stabilize the employment–wage dynamics of the growth cycle.

Overall, the literature supports treating labor force participation as an endogenous outcome shaped by income distribution, health behavior, and productivity dynamics, precisely as formalized in the extended Goodwin predator–prey framework with substance abuse.

2.3 A quick look at the data

This section provides a brief empirical overview of how labor share shocks are distributed across U.S. states and time, and how different levels of shock severity relate to drug-induced mortality. The analysis draws on a state-level panel covering all 50 U.S. states over 1998–2019. The outcome variable is the age-adjusted drug-induced mortality rate per 100,000 population. Labor share shocks are constructed as year-over-year changes in the state-level labor share, standardized by their cross-sectional standard deviation, and classified into nine bins: a no-shock category (Bin 0) for changes within ± 0.5 standard deviations, four positive shock bins (Bins 1–4) for increases of $+0.5$ to $+1.0$, $+1.0$ to $+1.5$, $+1.5$ to $+2.0$, and above $+2.0$ standard deviations, and four negative shock bins (Bins 5–8) for symmetric decreases. This classification captures both the direction and severity of labor share movements and enables a dose-response analysis across shock intensities.

Timing of labor share shocks. Figure 1 maps the timing of labor share shocks for each U.S. state across the full sample period. Each row corresponds to a state (sorted by the year of its first negative shock), and each cell is colored gray for no-shock years, navy for positive shocks, and red for negative shocks. Two broad patterns emerge. First, positive shocks are more prevalent in the late 1990s and early 2000s, coinciding with the expansion and tight labor markets of that era. Second, negative shocks become substantially more widespread after 2007–2008, reflecting the labor share compression associated with the Great Recession:

by 2009, the majority of states had experienced at least one negative labor share shock. A third feature is the recurring nature of treatment: most states appear in multiple shock categories across different years, alternating between positive and negative episodes within the sample window. This pattern of staggered and recurring shocks motivates the use of Teulings–Zubánov local projections rather than a standard single-event-study design.

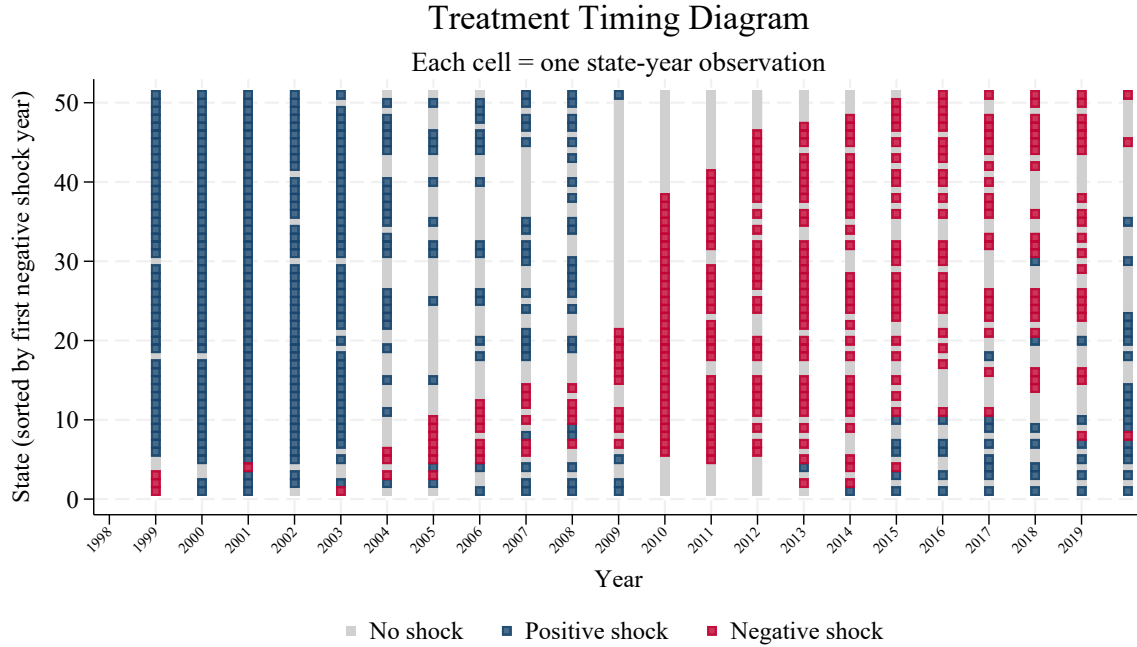


Figure 1: Treatment timing of labor share shocks across U.S. states, 1998–2019. Each row is a state, sorted by year of first negative shock. Gray cells indicate no-shock years (within ± 0.5 SD); navy cells are positive labor share shocks (Bins 1–4); red cells are negative labor share shocks (Bins 5–8). The concentration of red cells after 2008 reflects the widespread labor share compression associated with the Great Recession.

Static relationship between shock severity and drug mortality. Figure 2 plots mean residual log drug-induced mortality by shock bin, after removing state fixed effects, year fixed effects, and cyclical controls (manufacturing employment share and the unemployment rate). These residuals capture the within-state, within-year variation in drug mortality associated with each shock category, net of aggregate trends and observable confounders.

The figure reveals a broadly monotone dose-response relationship. States in the most severe negative shock category (Bin 8, labor share declines exceeding 2 standard deviations) exhibit residual drug mortality approximately 3.3 log points above the sample mean, while states experiencing moderate positive shocks (Bin 2, increases of 1.0–1.5 SD) show mortality roughly 1.0 log point below average. Intermediate negative shock bins (Bins 5–7) are associated with modestly positive residuals, while positive shock bins (Bins 1–3) lie below zero. A fitted quadratic curve captures the overall shape: drug mortality declines as shocks move from the most severe negative to the positive range, though the Bin 8 estimate carries little weight given only three treated state-year observations. This cross-bin pattern is consistent with the model’s central prediction that labor income compression raises the incidence of substance abuse, while labor income gains are protective.

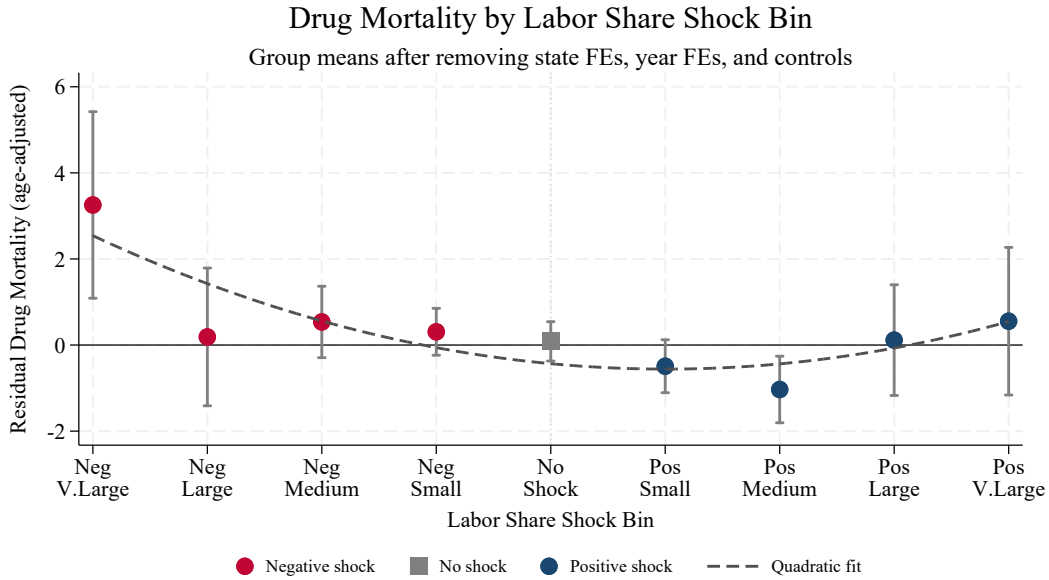


Figure 2: Mean residual log drug-induced mortality (age-adjusted) by labor share shock bin, after removing state fixed effects, year fixed effects, manufacturing employment share, and the unemployment rate. Bins 5–8 correspond to negative shocks of increasing severity; Bin 0 is the no-shock baseline; Bins 1–4 are positive shocks of increasing severity. The fitted curve is a quadratic polynomial.

Dynamic effects: local projection impulse responses. To assess the dynamic causal effects of labor share shocks on drug mortality, we estimate event-style local projections following [Teulings and Zubanov \(2014\)](#) for each shock bin separately. Each regression controls for state and year fixed effects, the full set of nuisance shock-bin indicators (all bins other than the focal bin), manufacturing employment share, and the unemployment rate. The outcome is the cumulative log change in age-adjusted drug mortality relative to the year before the shock ($t = -1$). Pre-period estimates ($t = -6$ to $t = -2$) test for differential pre-trends, while post-period estimates ($t = 0$ to $t = +5$) recover the dynamic causal effect.

Figure 3 presents the impulse responses for positive shock bins (Bins 1–4). Pre-period estimates are close to zero across all bins, supporting the parallel trends assumption. In the post-period, estimated effects are generally negative, consistent with positive labor share gains reducing drug mortality. Bin 2 (positive medium shocks, $n = 140$) shows the most consistent negative post-period response, reaching approximately -0.1 log points by horizons $+2$ to $+4$. Bin 4 (positive very large shocks, $n = 16$) exhibits a declining post-period trajectory of similar magnitude, though with wider confidence intervals. Bins 1 and 3 produce smaller post-period effects whose 95% confidence bands straddle zero.

Figure 4 presents the impulse responses for negative shock bins (Bins 5–8). The pattern is less uniform. Bin 5 (negative small shocks, $n = 214$) shows near-zero estimates throughout both the pre- and post-periods, with relatively narrow confidence bands, suggesting that small negative labor share shocks have little discernible dynamic effect in isolation. Bin 6 (negative medium shocks, $n = 85$) exhibits a similar near-null post-period pattern. Bin 7 (negative large shocks, $n = 14$) displays more volatile post-period estimates that rise briefly above zero at horizons $+1$ to $+2$ before returning toward zero, though wide confidence intervals preclude precise inference. Bin 8 (negative very large shocks, $n = 3$) is too sparsely identified to yield informative estimates: the 95% confidence bands span roughly -0.2 to $+0.8$ log points across the post-period.

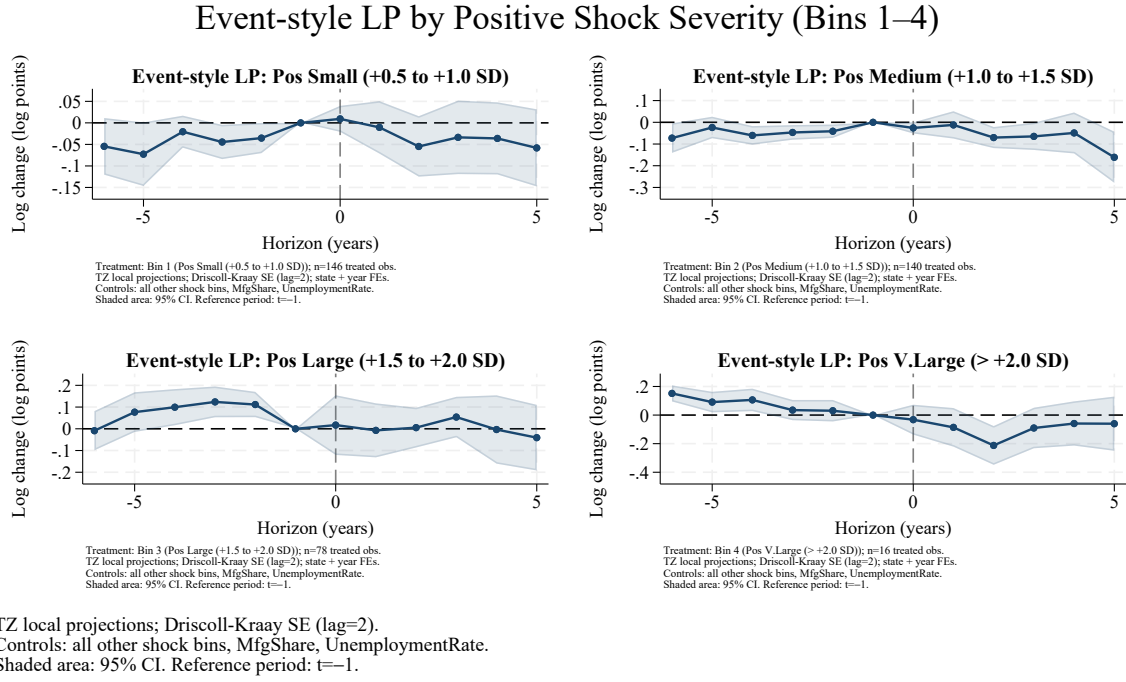


Figure 3: Event-style local projection impulse responses for positive labor share shock bins (Bins 1–4). The outcome is the cumulative log change in age-adjusted drug mortality relative to $t = -1$. Shaded areas are 95% confidence intervals based on Driscoll–Kraay standard errors (lag 2). All regressions include state and year fixed effects, manufacturing employment share, unemployment rate, and all other shock bins as nuisance controls. The vertical dashed line marks the shock year; the horizontal dashed line marks zero.

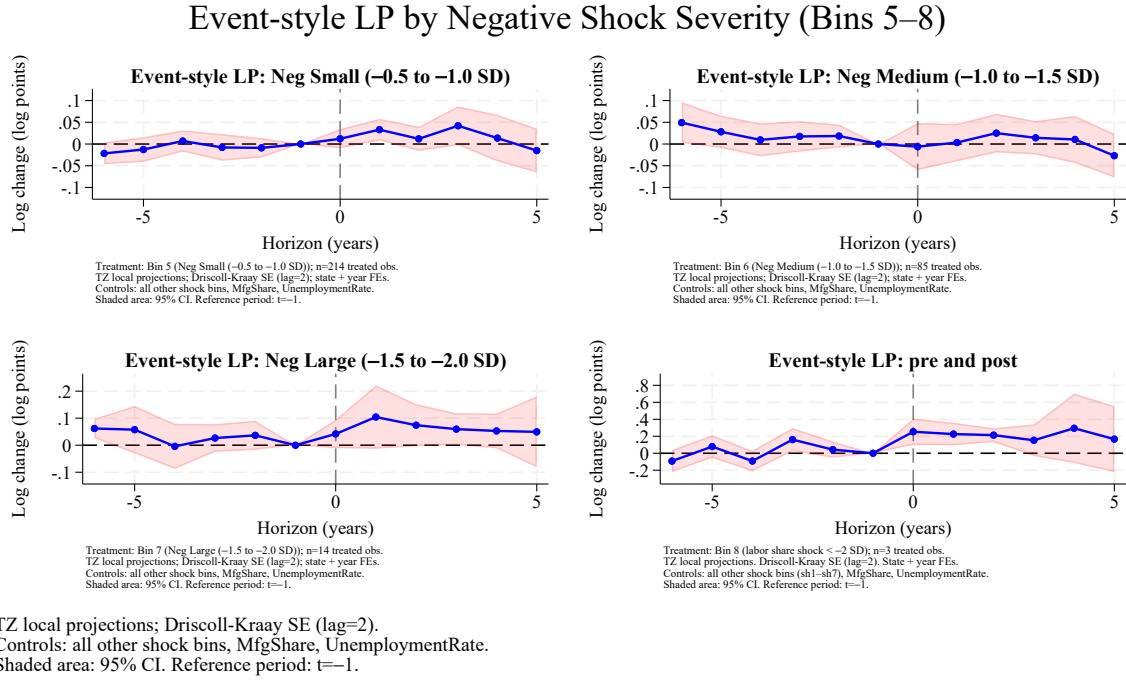


Figure 4: Event-style local projection impulse responses for negative labor share shock bins (Bins 5–8). The outcome is the cumulative log change in age-adjusted drug mortality relative to $t = -1$. Shaded areas are 95% confidence intervals based on Driscoll–Kraay standard errors (lag 2). All regressions include state and year fixed effects, manufacturing employment share, unemployment rate, and all other shock bins as nuisance controls. Bin 8 ($n = 3$) should be interpreted with considerable caution given the very small number of treated observations.

Taken together, the cross-bin and dynamic evidence point to an asymmetric relationship between labor share shocks and drug mortality. Positive labor share gains—particularly at moderate severity—are associated with statistically meaningful reductions in drug-induced deaths, consistent with the income-protective channel formalized in the model. Negative shocks, while associated with elevated drug mortality in the static cross-bin comparison, produce less precisely estimated dynamic effects, especially at severe levels where treated observations are scarce. The static group-means pattern and the local projection estimates together provide empirical grounding for the model’s core mechanism: that income distribution dynamics shape substance use behavior and, through this channel, aggregate health outcomes.

3 The model

[...] Opioid Use Disorder (OUD) [...]

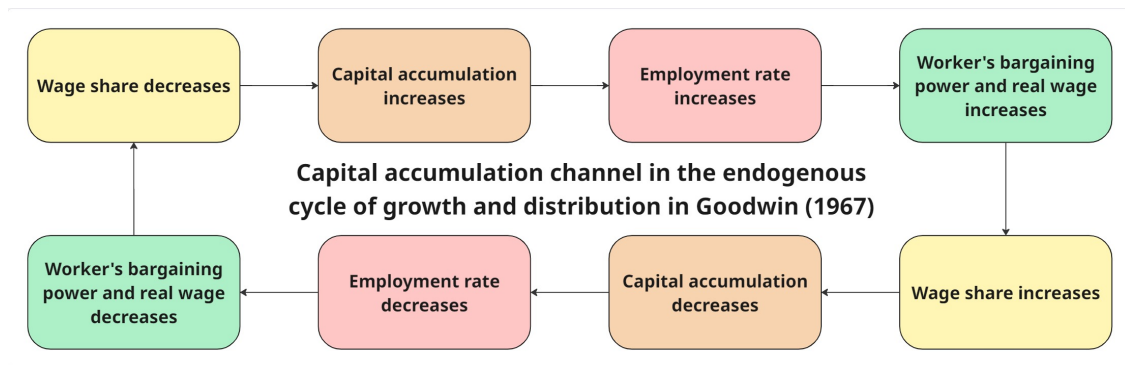


Figure 5: Capital accumulation channel in the endogenous cycle of growth and distribution in [Goodwin \(1967\)](#). A declining wage share raises the profit share, stimulating capital accumulation and employment. As the labor market tightens, workers’ bargaining power recovers, real wages rise, the wage share increases, and profits are squeezed, reversing the expansion. This two-variable predator–prey mechanism generates the closed orbits in the (e, u) -plane of the original model.

Before introducing the substance-abuse extension, it is useful to recall the core feedback loop underlying [Goodwin \(1967\)](#)’s original growth-cycle model. Figure 5 traces the capital accumulation channel that drives the endogenous oscillations between employment and income distribution. Starting from a phase of low wage share, high profitability stimulates investment and capital accumulation, which expands output and labor demand relative to the fixed workforce, pushing the employment rate upward. Rising employment tightens the labor market, strengthens workers’ bargaining power, and drives real wages above productivity growth, compressing the profit share. The resulting decline in accumulation reverses the employment expansion, completing the cycle. In the canonical two-dimensional system, the interplay between these two forces generates a conservative dynamical system whose orbits are entirely determined by initial conditions. The extended model developed in the following subsections breaks this conservative structure by introducing a third state variable—the workforce health composition index z —that interacts with both the accumulation and the wage-distribution dynamics.

3.1 Modeling abuse and non-abuse probabilities

Suppose the workforce (n) is constant and divided into two groups, those with opioid use disorder (n^{OUD}) and individuals non-OUD (n^{NOUD}), so that:

$$n = n^{OUD} + n^{NOUD}$$

The fractions of agents in each state evolve according to a continuous-time version of the discrete-choice approach (e.g. Brock and Hommes, 1997). We thus write:

$$\begin{aligned}\frac{\dot{n}^d}{n} &= p^d(U^d) - \frac{n^{OUD}}{n} \\ \frac{\dot{n}^c}{n} &= p^c(U^c) - \frac{n^{NOUD}}{n}\end{aligned}\tag{1}$$

where p^d and p^c are the probabilities of belonging to each group. They depend on observable factors comprised on U^d and U^c . In line with the discrete-choice literature (for a literature review see Franke and Westerhoff, 2017), we have:

$$\begin{aligned}p^d(U^d) &= \frac{\exp(U^d)}{\exp(U^d) + \exp(U^c)} \\ p^c(U^c) &= \frac{\exp(U^c)}{\exp(U^d) + \exp(U^c)}\end{aligned}\tag{2}$$

We assume that the same variables influence both groups with the opposite sign, i.e. $U^d = -U^c$. Empirical evidence indicates reductions in the wage share (u) are associated with a higher incidence of drug and alcohol abuse. On the contrary, a higher size of output going to workers increases the probability of being drug-clean. Therefore, we write:

$$U^d = -U^c = -\gamma u\tag{3}$$

where γ is frequently referred to as the intensity of choice parameter.

Substituting Eq. (3) into (2) and the resulting expressions into (1), we have:

$$\begin{aligned}\frac{\dot{n}^d}{n} &= \frac{\exp(-\gamma u)}{\exp(-\gamma u) + \exp(\gamma u)} - \frac{n^{OUD}}{n} \\ \frac{\dot{n}^c}{n} &= \frac{\exp(\gamma u)}{\exp(-\gamma u) + \exp(\gamma u)} - \frac{n^{NOUD}}{n}\end{aligned}\tag{4}$$

Subtracting the second from the first expression in (4), we obtain:

$$\dot{z} = \tanh(-\gamma u) - z\tag{5}$$

where

$$z = \frac{n^{OUD} - n^{NOUD}}{n} \in (-1, 1)$$

is an index capturing the composition of the population between those that drug abuse and drug-clean agents. When $z = 1$, all individuals are in the first group, while for $z = -1$, nobody abuses alcohol or drugs. An index $z = 0$ indicates the workforce is equally divided between both groups. It goes without saying that z can be at any point of the spectrum.

3.2 Introducing the prey

In Goodwin's (1967) original growth cycle model, the employment rate played the role of the "prey". Consider an economy that operates with a Leontief production technology:

$$q = \min \left\{ \frac{k}{v}; ena \right\}$$

where q stands for output, k is the capital stock, $v = k/q$ is the capital-output ratio, $e = l/n$ is the employment rate, with l as labor demand, and $a = q/l$ corresponds to labor productivity. Provided that v is constant and recalling our assumption of a constant population, from the Leontief dynamic efficiency condition follows that:

$$\begin{aligned} \frac{\dot{q}}{q} &= \frac{\dot{k}}{k} \\ \frac{\dot{e}}{e} &= \frac{\dot{q}}{q} - \frac{\dot{a}}{a} \end{aligned} \tag{6}$$

Substituting the first into the second expression in (6), we have:

$$\frac{\dot{e}}{e} = \frac{\dot{k}}{k} - \frac{\dot{a}}{a} \tag{7}$$

All wages are assumed to be consumed, while profits are saved and invested:

$$\dot{k} = q - wl = \left(1 - \frac{wl}{q}\right) q = (1 - u)q$$

where $u = wl/q = w/a$ is the share of workers in total output, i.e. the wage share. This implies:

$$\frac{\dot{k}}{k} = \frac{1 - u}{v} \tag{8}$$

Labor productivity is assumed to grow at an exogenous rate α , but discounted by the damage effects of drug abuse:

$$\frac{\dot{a}}{a} = [1 - D(z)]\alpha \tag{9}$$

with $D'(\cdot) > 0$ indicating that $D(\cdot)$ is an increasing function in its main argument. A higher share of individuals abusing the use of alcohol or drugs has damage impact on productivity.

Substituting Eqs. (8) and (9) into (7), we obtain the dynamics of the employment rate depending on income distribution and the composition of the population in terms of drug use:

$$\frac{\dot{e}}{e} = \frac{1 - u}{v} - [1 - D(z)]\alpha \tag{10}$$

Equation (10) reveals that employment dynamics are driven by two terms with opposite signs. The capital accumulation term $(1 - u)/v$ is the classical Goodwin channel. The second term, $[1 - D(z)]\alpha$, is the effective rate of labor productivity growth, which appears with a negative sign: faster productivity growth erodes employment by reducing the labor required to produce a given level of output. The substance-abuse channel depicted in Figure 6 operates precisely through this second term. When the wage share u falls, the population composition index z shifts toward higher OUD prevalence, raising $D(z)$ and thereby *reducing* effective productivity growth $[1 - D(z)]\alpha$. Because this term enters negatively, a smaller

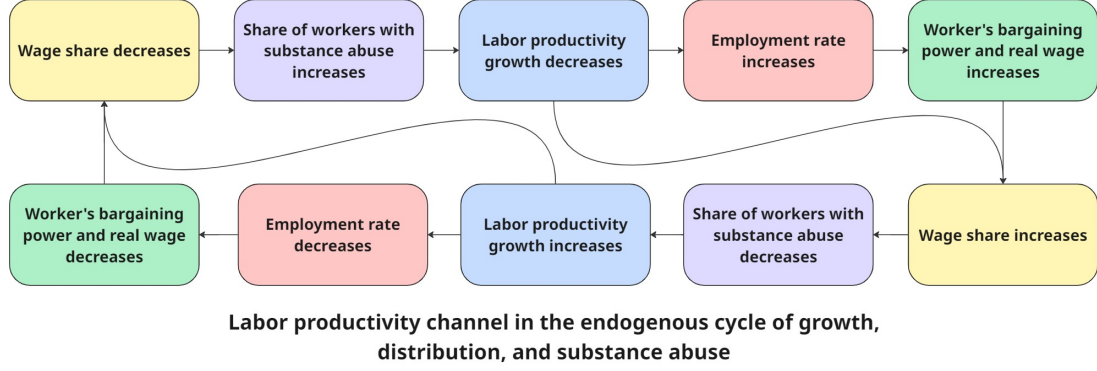


Figure 6: Labor productivity channel in the endogenous cycle of growth, distribution, and substance abuse. A declining wage share raises the incidence of substance use disorder among workers, which depresses the effective rate of labor productivity growth via the damage function $D(z)$. Lower productivity growth raises the employment rate based on the equation ((10)) while in the reverse phase a rising wage share reduces substance abuse, restores productivity growth, and eventually compresses employment. This channel is absent from Goodwin (1967)’s original formulation and constitutes the novel feedback introduced in the present model.

value means a weaker drag on employment: the economy requires more workers to sustain the same output when productivity growth is depressed. Substance abuse therefore acts as an *amplifier* of the accumulation-driven employment expansion—both forces simultaneously push $\dot{e}/e$ upward during the low-wage-share phase. In the reverse phase, when the wage share recovers, OUD prevalence declines, $D(z)$ falls, effective productivity growth rebounds, and the stronger productivity growth accelerates the contraction in employment. This amplification mechanism is what breaks the conservative structure of the original Goodwin model and, above the bifurcation threshold identified in Proposition 4, gives rise to self-sustaining limit cycles.

3.3 The predator arrives

Taking log-derivatives of the wage share definition, we have that:

$$\frac{\dot{u}}{u} = \frac{\dot{w}}{w} - \frac{\dot{q}}{q} \quad (11)$$

Assuming the existence of a real wages Phillips curve:

$$\frac{\dot{w}}{w} = f(e) \quad (12)$$

with $f'(e) > 0$ indicating that $f(\cdot)$ is an increasing function of the employment rate. As e increases, the bargaining power of workers improves, and they are able to obtain real wage increases.

Substituting Eqs. (9) and (12) into (11), we can write the dynamics of income distribution in terms of the evolution of employment rates and drug abuse:

$$\frac{\dot{u}}{u} = f(e) - [1 - D(z)]\alpha \quad (13)$$

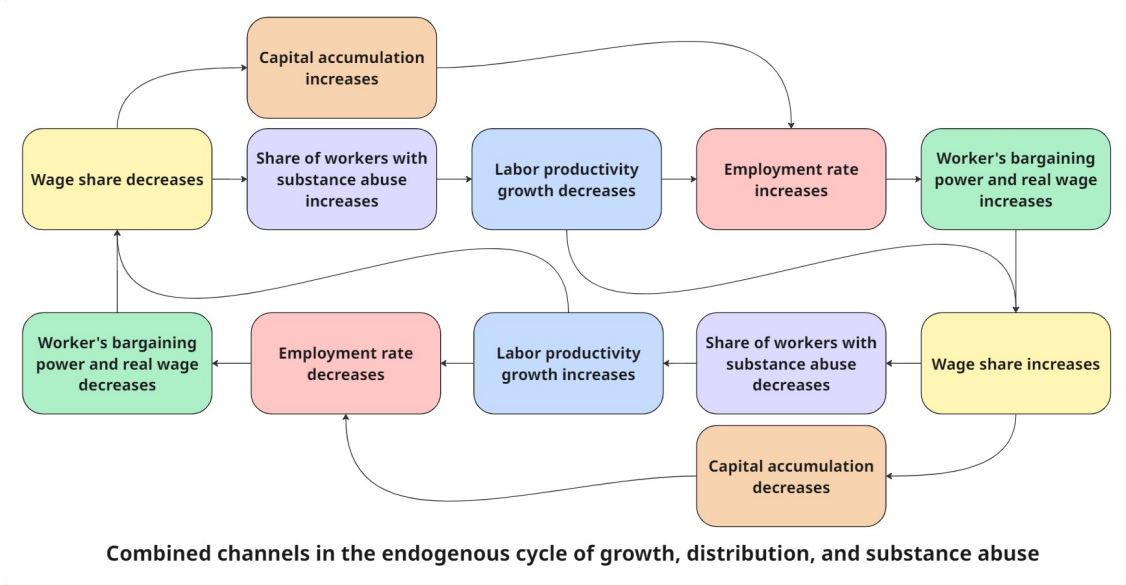


Figure 7: Combined channels in the endogenous cycle of growth, distribution, and substance abuse. Both the capital accumulation channel (upper path) and the labor productivity channel (middle path) are simultaneously active and operate in the same direction. Starting from a declining wage share, the profit share rises, stimulating capital accumulation (upper branch), while the incidence of substance use disorder simultaneously increases, depressing effective productivity growth (middle branch). Both effects independently raise the employment rate, amplifying the expansion phase of the cycle. In the contraction phase the roles are symmetrically reversed: a rising wage share simultaneously depresses capital accumulation and reduces substance abuse prevalence, with both forces jointly compressing the employment rate and propagating the next downswing. The three-dimensional system (14) formalizes this joint dynamics.

3.4 The dynamic system

Figure 7 synthesizes the two channels into the complete feedback structure of the model. At every point in the cycle, the wage share u simultaneously governs investment behavior—through the profit share $(1-u)$ and its effect on capital accumulation—and health behavior—through the discrete-choice mechanism that determines the workforce composition index z . These two channels reinforce each other in the employment expansion phase: rising profitability accelerates capital deepening while the associated decline in the wage share raises substance abuse prevalence and depresses productivity growth, each independently pushing the employment rate toward its cyclical peak. Conversely, in the contraction phase, the compression of profits slows accumulation while the improvement in the wage share reduces OUD prevalence and restores effective productivity growth, jointly dampening employment. The nonlinear interaction between these channels—mediated through the hyperbolic tangent function in equation (5) and the damage function $D(z)$ —is what gives rise to the rich dynamical behavior documented in Section 4, including the possibility of Hopf bifurcations and the coexistence of multiple stable limit cycles.

Our addicted predator–prey model is described by the following 3-dimensional nonlinear dynamic system:

$$\begin{aligned}\dot{e} &= \left[\frac{1-u}{v} - (1-D(z))\alpha \right] e \\ \dot{u} &= [f(e) - [(1-D(z))\alpha]u] \\ \dot{z} &= \tanh(-\gamma u) - z\end{aligned}\tag{14}$$

In equilibrium, $\dot{e} = \dot{u} = \dot{z} = 0$, giving us the following equilibrium conditions:

$$\begin{aligned}0 &= \left[\frac{1-\bar{u}}{v} - (1-D(\bar{z}))\alpha \right] \bar{e} \\ 0 &= [f(\bar{e}) - [(1-D(\bar{z}))\alpha]\bar{u}] \\ 0 &= \tanh(-\gamma\bar{u}) - \bar{z}\end{aligned}\tag{15}$$

Thus, we can state and prove the following Proposition regarding the existence of a unique or multiple equilibria.

Proposition 1 *The dynamic system (14) has a unique economically meaningful equilibrium point $P_1 = (\bar{e}_1, \bar{u}_1, \bar{z}_1)$ defined and given by:*

$$\begin{aligned}\bar{u} &= H^{-1}(1 - v\alpha) \\ \bar{z} &= \tanh(-\gamma H^{-1}(1 - v\alpha)) \\ \bar{e} &= f^{-1}(\alpha [1 - D(\tanh(-\gamma H^{-1}(1 - v\alpha)))])\end{aligned}$$

where

$$H(u) = u - \alpha v D(\tanh(-\gamma u))$$

Proof. See Appendix A.1. ■

Proposition 2 *The equilibrium point $P_1 = (\bar{e}_1, \bar{u}_1, \bar{z}_1)$ is locally asymptotically stable provided that:*

$$1 - ef' > 0$$

Moreover, when:

$$1 - ef' = 0$$

The equilibrium loses stability, and a Hopf bifurcation occurs.

Proof. See Appendix A.2. ■

To provide a more concrete view of the properties of the model, we adopt a set of functional forms for the behavioral equations of the model. This step allows us to restate the existence and local stability conditions of the equilibrium in terms of parameters that later on can be calibrated following empirical studies in the field and well-known time series:

$$\begin{aligned} D(z) &= \theta(1 + z) \\ f(e) &= -\beta_0 + \beta_1 e \end{aligned} \tag{16}$$

where θ is the sensitivity of damage in labor productivity to drug abuse; $-\beta_0$ is autonomous real wage rate growth; β_1 is the sensitivity of real wage rate growth to employment.

Proposition 3 *The dynamic system (14) admits a unique economically meaningful equilibrium solution that satisfies:*

$$\begin{aligned} \bar{u} &= J^{-1}(1) \\ \bar{z} &= \tanh(-\gamma J^{-1}(1)) \\ \bar{e} &= \frac{\beta_0 - \alpha \tanh(-\gamma J^{-1}(1))}{\beta_1} \end{aligned}$$

where

$$J(u) = u - \alpha v \tanh(-\gamma u)$$

Proof. See Appendix A.3. ■

Proposition 4 *For values of β_0 in the neighborhood of the critical value*

$$\beta_0 - \alpha \tanh(-\gamma J^{-1}(1)) = 1 \tag{17}$$

the dynamic system (15)–(17) has a family of periodic solutions.

Proof. See Appendix A.4. ■

4 Numerical simulations

The analytical results established in Section 3, the existence of a unique interior equilibrium (Propositions 1 and 3) and the conditions for local stability and Hopf bifurcation (Propositions 2 and 4), are now illustrated through a series of numerical experiments. Throughout this section we use the functional forms in equation (16) and calibrate the parameter values following established conventions in the Goodwin-cycle literature. Unless stated otherwise, the baseline parameters are $v = 6.25$, $\alpha = 0.05$, $\beta_1 = 1.67$, $\gamma = 1.96$, and $\theta = 2$.

Calibration follows the empirics of the U.S. economy, with parameters adjusted so that the model equilibrium $(\bar{e}, \bar{u})$ aligns with observed values of the employment-population ratio and the labor share of national income. The capital-output ratio is set to $v = 6.25$ and the exogenous growth rate of labor productivity to $\alpha = 0.05$. The Phillips curve parameters

Table 1: Parameter values

Parameter	Value	Description	Source
v	6.25	Capital-output ratio	to make $\bar{e} = 0.6$ and $\bar{u} = 0.75$
α	0.05	Labor productivity growth rate	to make $\bar{e} = 0.6$ and $\bar{u} = 0.75$
β_0	0.96	Autonomous real wage growth	Hopf bifurcation condition
β_1	1.67	Wage sensitivity to employment	Hopf bifurcation condition
γ	1.96	Intensity of choice parameter	Bifurcation condition
θ	2.00	Productivity damage parameter	Bifurcation condition

$\beta_0 = 0.96$ and $\beta_1 = 1.67$ are jointly chosen to satisfy the Hopf bifurcation condition established in Proposition 2, ensuring that the calibrated equilibrium lies in the neighborhood of the critical value at which persistent cyclical fluctuations emerge. The intensity of choice parameter $\gamma = 1.96$, governing the sensitivity of substance abuse probabilities to income distribution, and the productivity damage parameter $\theta = 2$, capturing the responsiveness of labor productivity to OUD prevalence, are selected to yield a damage function $D(z)$ compatible with the bifurcation condition while preserving economically meaningful dynamics. Table 1 summarizes the parameter values.

4.1 Convergence to equilibrium below the bifurcation threshold

We begin by documenting the dynamical behavior of system (14) in the stable regime, that is, for parameter values that satisfy the local stability condition of Proposition 2, $1 - \bar{e}f'(\bar{e}) > 0$. Setting $\beta_0 = 0.6$, the Hopf bifurcation condition $\bar{e}f'(\bar{e}) = 1$ is not met, and the unique interior equilibrium $P_1 = (\bar{e}, \bar{u}, \bar{z})$ is locally asymptotically stable. The equilibrium coordinates under this parametrization are approximately $\bar{e} \approx 0.38$, $\bar{u} \approx 0.75$, and $\bar{z} \approx -0.90$, indicating that the long-run composition of the workforce is heavily skewed toward drug-clean individuals, consistent with a relatively moderate wage share.

Figures 8 and 9 report the numerical results for two trajectories starting from different initial conditions: the blue trajectory is initialized at $(\bar{e}, \bar{u} + 0.03, \bar{z})$, perturbing only the wage share, while the red trajectory is initialized at $(\bar{e} + 0.03, \bar{u}, \bar{z})$, perturbing only the employment rate, both by the same magnitude of 0.03.

Figure 8 displays the phase portrait in the (e, u, z) -space. Both trajectories trace inward-spiraling paths that progressively tighten around the equilibrium point P_1 . The red trajectory, initialized with a positive displacement in the employment rate $(\bar{e} + 0.03, \bar{u}, \bar{z})$, executes wide oscillatory loops before gradually contracting toward the interior of the attractor. The blue trajectory, displaced instead along the wage-share dimension $(\bar{e}, \bar{u} + 0.03, \bar{z})$, spirals inward along a distinct path, reflecting the asymmetric role that e and u play in the system's dynamics. Crucially, the two paths converge to the same fixed point regardless of the initial condition, confirming the global attracting character of the equilibrium within the region explored. This result stands in sharp contrast to the original Goodwin (1967) model, which is a conservative dynamical system: in the canonical two-dimensional case, every trajectory constitutes a closed orbit determined entirely by initial conditions, and no path converges to the interior equilibrium. The introduction of the health-deterioration channel through the population composition variable z breaks this conservative structure and can generate genuine asymptotic stability when the autonomous wage-growth parameter β_0 lies below the bifurcation threshold.

The time-series representation in Figure 9 complements this picture by tracing the evolution of each state variable individually. All three panels display damped oscillatory dynam-

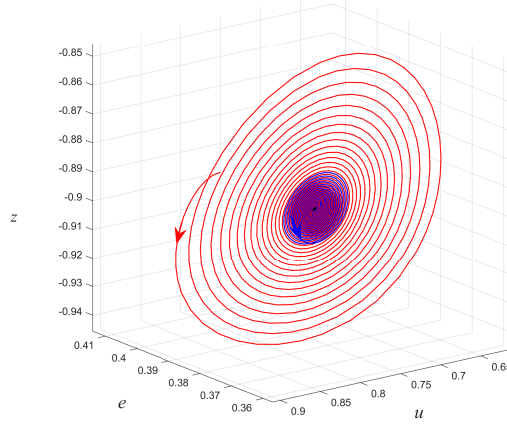


Figure 8: Convergence to the equilibrium in the three-dimensional state space (e, u, z) . Initial conditions for the blue and red trajectories are $(\bar{e}, \bar{u} + 0.03, \bar{z})$ and $(\bar{e} + 0.03, \bar{u}, \bar{z})$, respectively. Both spiral inward and converge to the equilibrium point (black dot). Parameters: $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.6$, $\beta_1 = 1.67$, $\gamma = 1.96$, and $\theta = 2$.

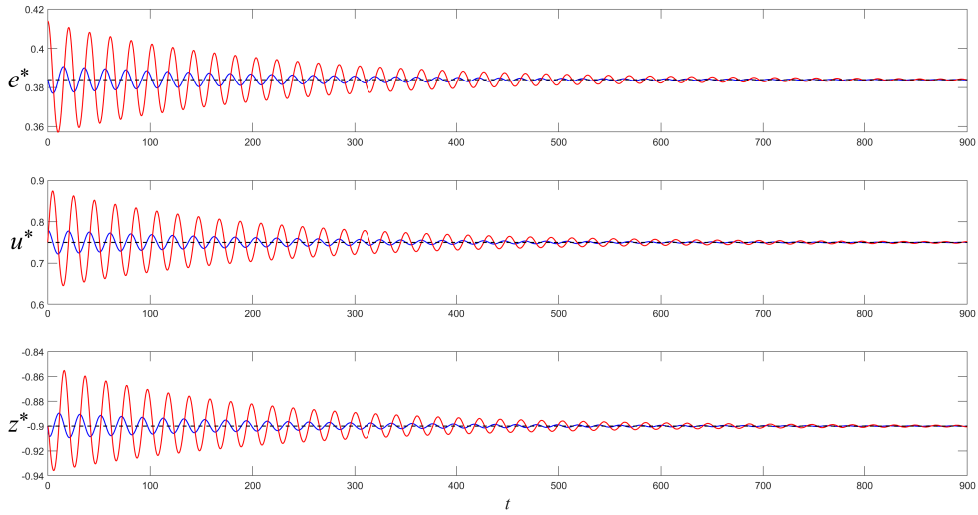


Figure 9: Time-series trajectories of the employment rate (e^* , top panel), the wage share (u^* , middle panel), and the population composition index (z^* , bottom panel). Initial conditions for the blue and red trajectories are $(\bar{e}, \bar{u} + 0.03, \bar{z})$ and $(\bar{e} + 0.03, \bar{u}, \bar{z})$, respectively. All three variables exhibit damped oscillations that converge to their respective steady-state values as $t \rightarrow \infty$. Parameters as in Figure 8.

ics: the amplitude of the fluctuations in the employment rate e^* , the wage share u^* , and the drug-abuse index z^* declines monotonically over time, with both trajectories converging to their respective steady-state values well before $t = 900$. The red trajectory, originating from a displacement in employment $(\bar{e} + 0.03, \bar{u}, \bar{z})$, exhibits noticeably larger amplitude in the early periods across all three panels, while the blue trajectory, displaced in the wage share $(\bar{e}, \bar{u} + 0.03, \bar{z})$, shows a comparatively milder transient response. Despite the difference in initial conditions, both trajectories eventually merge and asymptote to the same equilibrium, confirming that the perturbation direction does not affect the long-run outcome. The oscillatory structure of the convergence path reflects the complex conjugate eigenvalues of the Jacobian at P_1 : the characteristic equation possesses one negative real root and a pair of complex roots with strictly negative real part, so the approach to equilibrium takes the form of a damped spiral in all three dimensions simultaneously.

Taken together, Figures 8 and 9 provide numerical confirmation of Proposition 2. They also illustrate a key economic implication: when the behavioral health channel is present but autonomous wage-growth pressure remains moderate ($\beta_0 = 0.6$), the feedback between income distribution, substance abuse, and productivity does not generate self-sustaining cyclical fluctuations. Instead, any shock to employment, the wage share, or workforce health composition is gradually absorbed, and the economy returns to a well-defined long-run steady state. As β_0 increases toward the critical bifurcation value β_0^{HB} identified in Proposition 4, this damping mechanism weakens — the real part of the complex eigenvalue pair approaches zero — and the equilibrium eventually loses stability in favor of persistent endogenous fluctuations. The two scenarios together delineate the boundary between a regime of macroeconomic resilience, in which health shocks are transitory, and a regime of endogenous instability, in which the interaction between substance abuse and income distribution perpetuates the growth cycle.

4.2 Coexistence of limit cycles above the bifurcation threshold

Having documented the stable regime in Section 4.1, we now turn to the dynamical behavior of system (14) when the Hopf bifurcation condition $\bar{e}f'(\bar{e}) = 1$ is crossed. Setting $\beta_0 = 0.96$, the bifurcation condition in Proposition 4 is satisfied, the interior equilibrium P_1 loses its local asymptotic stability, and the system admits endogenous, persistent oscillations. Figures 10 and 11 illustrate a qualitatively richer phenomenon: depending on the initial conditions, the economy does not converge to a single attracting cycle but instead settles on one of *three distinct stable limit cycles* that coexist in the state space (e, u, z) .

Figure 10 presents the asymptotic trajectories in the (e, u, z) -space, plotted over the late portion of the simulation after all transients have elapsed. Three clearly separated closed orbits are visible: an innermost cycle traced by the blue and green trajectories (nearly indistinguishable), an intermediate cycle traced by the red trajectories, and a large-amplitude outermost cycle traced by the black trajectories. The unstable equilibrium P_1 sits at the common centre of all three orbits. Each orbit is a genuinely closed, self-sustaining curve: trajectories initiated within the basin of attraction of a given cycle remain on that cycle indefinitely, neither converging inward toward the equilibrium nor diverging outward. The amplitude of each cycle — measured by the range of fluctuations in e , u , and z — increases systematically from the innermost to the outermost orbit, with the black cycle spanning substantially wider ranges than the blue and green cycles along every state-variable dimension.

A particularly instructive feature of Figure 10 is the near coincidence of the blue and green orbits despite their markedly different initial conditions. The blue trajectories are launched from small, single-axis perturbations of magnitude 0.01 around the equilibrium, whereas the green trajectories involve a substantially larger displacement of +0.05 in the

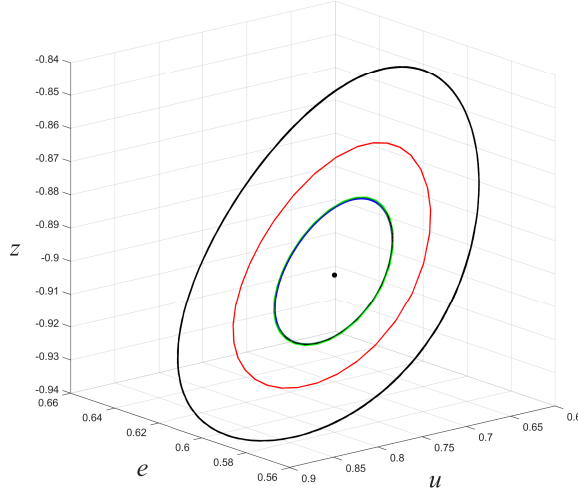


Figure 10: Coexistence of three limit cycles in the three-dimensional state space (e, u, z) , plotted over the asymptotic portion of the simulation. Initial conditions for the blue trajectories are $(\bar{e} + 0.01, \bar{u}, \bar{z})$, $(\bar{e}, \bar{u} + 0.01, \bar{z})$, and $(\bar{e}, \bar{u}, \bar{z} + 0.01)$. Initial conditions for the red trajectories are $(\bar{e} - 0.03, \bar{u}, \bar{z})$, $(\bar{e}, \bar{u} - 0.03, \bar{z})$, $(\bar{e}, \bar{u}, \bar{z} - 0.03)$, $(\bar{e} + 0.03, \bar{u}, \bar{z})$, $(\bar{e}, \bar{u} + 0.03, \bar{z})$, and $(\bar{e}, \bar{u}, \bar{z} + 0.03)$. Initial conditions for the black trajectories are $(\bar{e} - 0.04, \bar{u}, \bar{z})$, $(\bar{e}, \bar{u} - 0.04, \bar{z})$, and $(\bar{e}, \bar{u}, \bar{z} - 0.04)$. Initial conditions for the green trajectories are $(\bar{e} - 0.01, \bar{u} + 0.05, \bar{z})$, $(\bar{e} + 0.01, \bar{u} + 0.05, \bar{z})$, and $(\bar{e}, \bar{u} + 0.05, \bar{z} - 0.01)$. The unstable equilibrium P_1 is indicated by the black dot. Parameters: $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.67$, $\gamma = 1.96$, and $\theta = 2$.

wage share together with small perturbations in the employment rate. Yet both groups converge to the same innermost limit cycle. This indicates that the boundary separating the basins of attraction of the three coexisting cycles is not simply determined by the scalar distance from the equilibrium: the *direction* of the initial displacement matters. A large perturbation concentrated in the wage share, as for the green trajectories, need not lead to a higher-amplitude cycle, whereas a smaller perturbation in the employment rate or z index can be sufficient to place the trajectory in the basin of a larger orbit.

This finding extends the result of Proposition 4 beyond the local neighborhood of the bifurcation. While the Hopf bifurcation theorem guarantees the existence of a family of periodic orbits for parameter values in the neighborhood of the critical value β_0^{HB} , the numerical evidence establishes that, for $\beta_0 = 0.96$, the system supports multiple stable limit cycles simultaneously. The coexistence of these cycles implies that the long-run behavior of the economy is *history dependent*: the amplitude of the endogenous growth cycle to which the economy converges is determined by the size and direction of initial conditions rather than by parameters alone. Economies that begin with large deviations in the employment rate — corresponding to the black trajectories — lock into high-amplitude oscillations with wide swings in both the wage share and the drug-abuse index, whereas economies starting closer to the equilibrium settle into milder cycles of smaller amplitude.

Figure 11 confirms the persistent nature of these oscillations through the time-series representation evaluated over $t \in [149,900, 150,000]$, a horizon sufficiently long for all initial transients to have elapsed. All three panels show undamped, regular oscillations with constant amplitude, providing unambiguous numerical evidence that the three orbits in Figure 10 are genuine limit cycles rather than long-lived transients. The amplitude ordering is preserved across all three state variables: the black trajectories exhibit the widest fluctuations, while the blue and green trajectories, which nearly coincide, oscillate within a

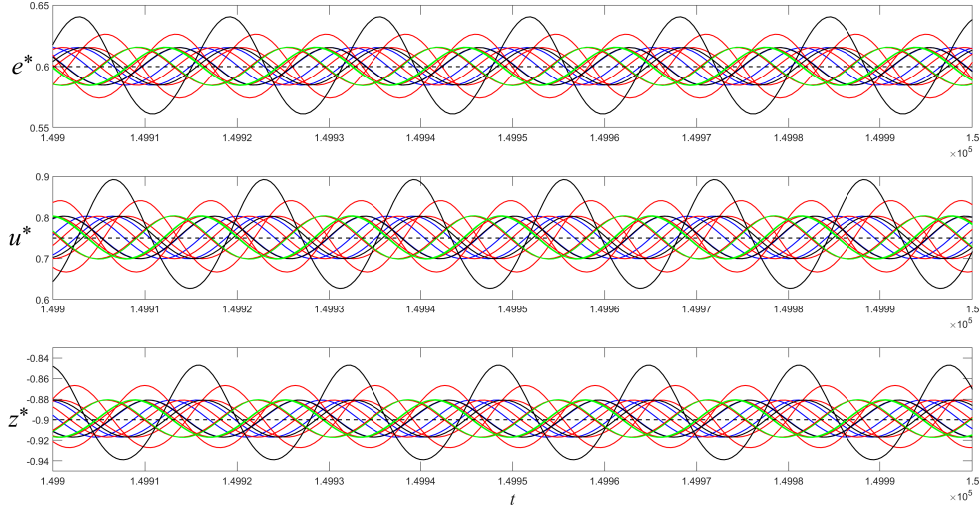


Figure 11: Time-series of the employment rate (e^* , top panel), the wage share (u^* , middle panel), and the population composition index (z^* , bottom panel) for the fifteen trajectories described in Figure 10, evaluated over $t \in [149900, 150000]$. Colors correspond to Figure 10: black (outermost cycle), red (intermediate cycle), and blue/green (innermost cycle, nearly coincident). The dashed line marks the unstable equilibrium value of each variable. Parameters as in Figure 10.

substantially narrower band around the equilibrium. The red trajectories occupy an intermediate position. Consistent with the basin-structure result discussed above, the green and blue time series are virtually identical despite originating from initial conditions of very different magnitudes, further confirming their convergence to the same innermost limit cycle. Notably, the oscillation frequencies across the three cycles are similar, reflecting that the period of the limit cycle is primarily governed by the eigenvalue structure of the Jacobian at P_1 , while the amplitude is determined by the size and direction of the initial displacement.

From an economic standpoint, the coexistence of multiple stable limit cycles carries a stark implication for the relationship between substance abuse and macroeconomic instability. Even holding all structural parameters of the economy constant — technology, the capital-output ratio, the wage Phillips curve — the severity of the growth cycle is not uniquely pinned down. Economies or historical episodes characterized by larger initial imbalances in the employment rate will be drawn into high-amplitude cycles in which the wage share, employment, and workforce health conditions fluctuate more violently. In such episodes, the drug-abuse index z^* undergoes larger swings, implying greater variation in the effective productivity growth rate $[1 - D(z)]\alpha$ over the cycle. This amplification mechanism can thus translate initial macroeconomic conditions into persistent differences in the magnitude of health-productivity feedbacks, even when the underlying behavioral and technological parameters are identical across economies.

4.3 A comparison with the original Goodwin growth cycle

Before turning to comparative dynamics, it is instructive to examine how the equilibrium of the extended model is determined and how it relates to the equilibrium of the original Goodwin (1967) model. In the canonical two-dimensional Goodwin model, the unique

interior equilibrium is given by:

$$\bar{u}^G = 1 - v\alpha, \quad \bar{e}^G = f^{-1}(\alpha), \quad (18)$$

where labor productivity grows at the exogenous rate α without any health-related discount. In the extended model, the presence of the drug-abuse channel modifies both equilibrium values: substance abuse reduces effective productivity growth to $[1 - D(\bar{z})]\alpha$, which shifts the equilibrium wage share and employment rate relative to their Goodwin counterparts. The drug-abuse index $\bar{z}$, in turn, is endogenously determined by the equilibrium wage share $\bar{u}$ through the discrete-choice mechanism in equation (5). This interdependence requires a two-step graphical procedure to locate the equilibrium, as illustrated in Figure 12.

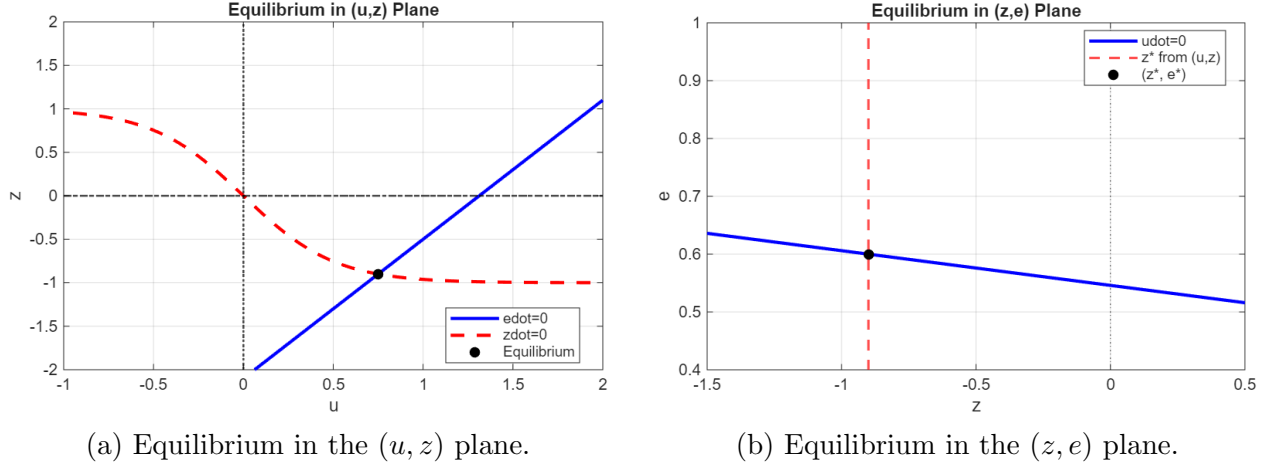


Figure 12: Two-step graphical determination of the interior equilibrium $P_1 = (\bar{e}, \bar{u}, \bar{z})$. Panel (a): the solid blue line is the $\dot{e} = 0$ nullcline and the dashed red curve is the $\dot{z} = 0$ nullcline in the (u, z) plane; their intersection (filled circle) yields $(\bar{u}, \bar{z})$. Panel (b): the solid blue line is the $\dot{u} = 0$ locus in the (z, e) plane; the dashed red vertical line carries forward $\bar{z}$ from panel (a), and their intersection (filled circle) yields $\bar{e}$. Parameters: $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.67$, $\gamma = 1.96$, and $\theta = 2$.

Step 1: Determination of $\bar{u}$ and $\bar{z}$. Panel (a) of Figure 12 plots the two nullclines that govern the joint determination of the equilibrium wage share and population composition. The $\dot{e} = 0$ locus (solid blue) follows from the employment-rate equation, which under the functional forms in equation (16) becomes a straight line with positive slope in the (u, z) plane:

$$z = -\frac{1}{\theta} \left[\frac{1-u}{v\alpha} - 1 \right] - 1.$$

The $\dot{z} = 0$ locus (dashed red) is the hyperbolic tangent function $z = \tanh(-\gamma u)$, which is strictly decreasing and bounded within $(-1, 1)$. Since the $\dot{e} = 0$ line is monotonically increasing and the $\dot{z} = 0$ curve is monotonically decreasing, the two loci cross exactly once in the economically relevant region, confirming the uniqueness result of Proposition 3. The intersection yields $\bar{u} \approx 0.75$ and $\bar{z} \approx -0.90$. The strongly negative value of $\bar{z}$ indicates that, at the equilibrium wage share, the workforce is heavily skewed toward drug-clean individuals: the discrete-choice mechanism implies that a wage share of $\bar{u} \approx 0.75$ provides sufficient labor income to keep the share of OUD workers well below the neutral point $z = 0$.

Step 2: Determination of $\bar{e}$. Given $\bar{z}$ from Step 1, the equilibrium employment rate is recovered from the $\dot{u} = 0$ condition, which pins down the level of e that makes real wage growth equal to effective productivity growth. Panel (b) of Figure 12 illustrates this step. The $\dot{u} = 0$ locus (solid blue) is a downward-sloping straight line in the (z, e) plane:

$$e = \frac{\beta_0 + \alpha(1 - \theta - \theta z)}{\beta_1},$$

which is decreasing in z because a higher drug-abuse index reduces effective productivity growth and thus the level of employment required to satisfy the real-wage Phillips curve. The dashed red vertical line carries forward the value $\bar{z} \approx -0.90$ obtained in Step 1, and its intersection with the $\dot{u} = 0$ locus yields $\bar{e} \approx 0.60$.

Comparison with the original Goodwin equilibrium. The two-step procedure highlights how the drug-abuse channel alters the long-run equilibrium relative to the original Goodwin model. Setting $\theta = 0$ — so that the damage function vanishes, $D(z) = 0$, and the health channel is switched off — the equilibrium wage share collapses to $\bar{u}^G = 1 - v\alpha = 1 - 6.25 \times 0.05 = 0.6875$, and the equilibrium employment rate becomes $\bar{e}^G = (\beta_0 + \alpha)/\beta_1 = (0.96 + 0.05)/1.67 \approx 0.605$. With $\theta = 2$, the extended model instead yields $\bar{u} \approx 0.75$ and $\bar{e} \approx 0.60$. The presence of substance abuse therefore raises the equilibrium wage share above its Goodwin benchmark: because OUD depresses effective productivity growth, the economy requires a lower profit share — and hence a higher wage share — to sustain the same capital-accumulation rate. The employment rate, by contrast, is only marginally affected by the health channel at the equilibrium, since the damage function is evaluated at the equilibrium drug-abuse index $\bar{z}$, which is close to -1 (nearly all workers are drug-clean). The key effect of the health channel operates through the *dynamics* around the equilibrium, reshaping the cycle rather than substantially displacing the steady state itself.

4.4 Sensitivity of OUD to income distribution

The intensity of choice parameter γ governs how strongly the wage share influences workers' transition probabilities between OUD and non-OUD states. Formally, γ enters the utility differential $U^d = -\gamma u$ in the discrete-choice mechanism of equation (5): a higher γ implies that a given decline in the wage share induces a proportionally larger increase in the probability of substance abuse, while a higher wage share more forcefully shifts workers toward the drug-clean state. In this sense, γ captures the *income-sensitivity of health behavior* — the degree to which substance-use decisions are responsive to economic conditions.

To situate the role of γ within the full feedback structure of the model, Figure 7 maps the combined channels linking income distribution, substance abuse, labor productivity, employment, and capital accumulation across both phases of the cycle.

Figure 13 compares the asymptotic limit cycles of the system under two values of γ : the baseline $\gamma = 1.96$ (blue) and a reduced value $\gamma = 1.96 \times 0.9 \approx 1.77$ (red), holding all other parameters fixed at $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.6$, and $\theta = 2$. For each parameterization, 15 initial conditions are simulated over $t \in [0, 150,000]$, each displacing the respective equilibrium by ± 0.01 , ± 0.03 , -0.04 , and mixed perturbations along individual state-variable dimensions. The orbits shown are extracted from the asymptotic portion of each simulation, ensuring that all transients have elapsed.

Two effects of reducing γ are immediately visible in Figure 13. First, the equilibrium shifts. Under the baseline $\gamma = 1.96$, the interior equilibrium is $(\bar{e}, \bar{u}, \bar{z}) = (0.600, 0.750, -0.900)$. Reducing γ to 1.96×0.9 moves the equilibrium to $(\bar{e}_1, \bar{u}_1, \bar{z}_1) = (0.598, 0.766, -0.875)$: the

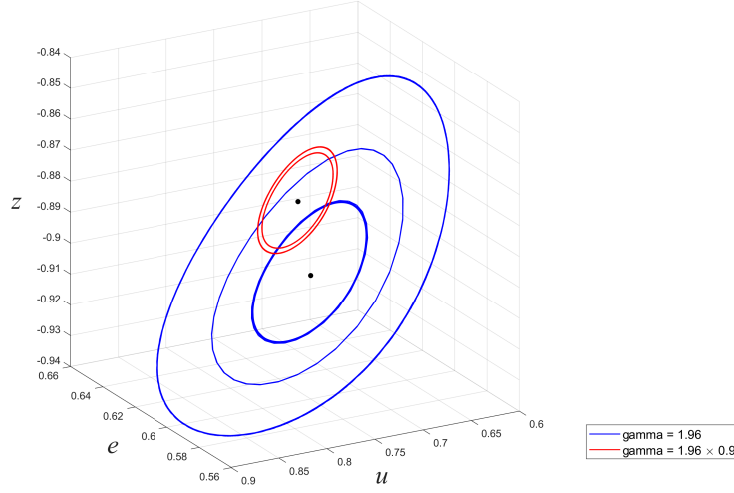


Figure 13: Effect of reducing the intensity of choice parameter γ on the limit cycles in (e, u, z) -space. Blue orbits: baseline $\gamma = 1.96$, with 15 trajectories initialized by displacing $(\bar{e}, \bar{u}, \bar{z})$ by ± 0.01 , ± 0.03 , -0.04 , and mixed perturbations along individual state-variable dimensions. Red orbits: $\gamma = 1.96 \times 0.9 \approx 1.77$, with 15 trajectories initialized by displacing the corresponding equilibrium $(\bar{e}_1, \bar{u}_1, \bar{z}_1)$ by the same perturbation patterns. Black dots indicate the respective equilibria P_1 for each parameterization. Other parameters: $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.\bar{6}$, and $\theta = 2$.

drug-abuse index rises from -0.900 to -0.875 (a less drug-clean workforce), the wage share rises from 0.750 to 0.766 , and the employment rate barely changes. The causal chain runs as follows. A smaller γ flattens the $\dot{z} = 0$ locus $\bar{z} = \tanh(-\gamma\bar{u})$, so that for any given wage share workers respond less strongly to income in their substance-use decisions and $\bar{z}$ becomes less negative. The resulting rise in OUD prevalence amplifies the productivity damage $D(\bar{z}) = \theta(1 + \bar{z})$, which lowers effective productivity growth $[1 - D(\bar{z})]\alpha$. To restore the $\dot{e} = 0$ condition $(1 - \bar{u})/v = [1 - D(\bar{z})]\alpha$, the economy requires a higher wage share $\bar{u}$ — consistent with the observed increase from 0.750 to 0.766 . The near-unchanged employment rate $\bar{e}$ follows directly from the $\dot{u} = 0$ condition, which pins down $\bar{e} = [\beta_0 + \alpha(1 - \theta(1 + \bar{z}))]/\beta_1$ independently of $\bar{u}$. The sensitivity of $\bar{e}$ to $\bar{z}$ is $-\alpha\theta/\beta_1 = -0.06$, and the shift in $\bar{z}$ is only 0.025 , yielding $\Delta\bar{e} \approx -0.06 \times 0.025 \approx -0.002$ — consistent with the observed decline from 0.600 to 0.598 .

Second, and more strikingly, a 10% reduction in γ dramatically compresses the amplitude of the limit cycles. The red orbits are substantially smaller than the blue orbits along every dimension of the state space. This compression reflects the weakening of the health channel's entry point, as Figure 7 makes explicit. Under high γ , a downturn in the wage share triggers a pronounced increase in OUD prevalence, which discounts productivity growth through the damage function $D(z)$, which in turn raises employment and reinforces the wage-share movement — a destabilizing spiral that sustains wide oscillations. When γ is lower, the same decline in u produces a smaller change in z , attenuating the health-productivity feedback and yielding narrower fluctuations. Importantly, because each parameterization's initial conditions are displaced from its own equilibrium by the same relative magnitudes, the striking difference in cycle amplitude between blue and red is attributable entirely to the change in γ rather than to differences in the initial displacement.

From an economic policy perspective, γ is not a purely structural parameter but one that reflects institutional and social conditions governing the link between economic stress

and substance abuse. Factors that effectively reduce γ include the generosity of social safety nets, universal access to healthcare and mental health services, housing security, and the availability of non-pharmacological coping mechanisms. When these buffers are in place, workers' substance-use behavior becomes less sensitive to cyclical income fluctuations, weakening the amplification channel between income distribution and health. The model thus suggests that social policy — by lowering γ — can play a macroeconomic stabilizing role above and beyond its direct health benefits, reducing both the amplitude of the growth cycle and the endogenous variation in workforce health composition over the business cycle.

4.5 Responsiveness of labor productivity to OUD

The parameter θ governs the magnitude of the productivity damage that substance abuse inflicts on the economy. Under the functional form $D(z) = \theta(1 + z)$, each unit increase in the OUD-prevalence index z reduces effective productivity growth by $\theta\alpha$. A higher θ therefore corresponds to an economy in which OUD workers impose a heavier toll on aggregate productivity — reflecting, for instance, a higher concentration of employment in physically demanding occupations susceptible to opioid-related impairment (Lundstrom et al., 2025), less access to workplace treatment accommodation, or more severe forms of addiction with greater presenteeism and absenteeism costs.

Figure 14 compares the asymptotic limit cycles under the baseline $\theta = 2$ (blue) and a 50% increase to $\theta = 3$ (red), holding all other parameters fixed at $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.6$, and $\gamma = 1.96$. For each parameterization, 15 initial conditions are simulated over $t \in [0, 200,000]$ — a longer horizon than the preceding experiments, reflecting the slower convergence to the limit cycle under higher productivity damage — with each trajectory initialized by displacing the respective equilibrium by ± 0.01 , ± 0.03 , -0.04 , and mixed perturbations. The orbits are again extracted from the asymptotic portion of the simulation.

Figure 14 reveals two distinct effects of raising θ . First, the equilibrium shifts. Under the baseline $\theta = 2$, the interior equilibrium is $(\bar{e}, \bar{u}, \bar{z}) = (0.600, 0.750, -0.900)$. Raising θ to 3 moves the equilibrium to $(\bar{e}_1, \bar{u}_1, \bar{z}_1) = (0.598, 0.773, -0.908)$: the wage share rises by 0.023, the drug-abuse index falls modestly from -0.900 to -0.908 (a marginally more drug-clean workforce), and the employment rate is nearly unchanged. The mechanism follows from Proposition 3: a higher θ amplifies the productivity cost of any given level of OUD prevalence, lowering effective productivity growth $[1 - D(\bar{z})]\alpha$. To restore the $\dot{e} = 0$ condition $(1 - \bar{u})/v = [1 - D(\bar{z})]\alpha$, the economy must sustain a higher wage share $\bar{u}$ — consistent with the observed rise from 0.750 to 0.773. Since $\bar{z} = \tanh(-\gamma\bar{u})$, the higher $\bar{u}$ in turn pushes $\bar{z}$ slightly more negative, from -0.900 to -0.908 , reflecting the self-correcting tendency toward a more drug-clean workforce when health costs are severe. The near-unchanged employment rate $\bar{e}$ again follows from the $\dot{u} = 0$ condition: $\bar{e} = [\beta_0 + \alpha(1 - \theta(1 + \bar{z}))]/\beta_1$, which is independent of $\bar{u}$. Here both θ and $\bar{z}$ change, so the relevant quantity is the product $\theta(1 + \bar{z})$, which shifts from $2 \times 0.100 = 0.200$ to $3 \times 0.092 = 0.275$. Multiplying the change of 0.075 by $\alpha/\beta_1 = 0.03$ yields $\Delta\bar{e} \approx -0.002$ — consistent with the observed decline from 0.600 to 0.598.

Second, and perhaps counterintuitively, the stronger productivity damage parameter *compresses* the amplitude of the limit cycles: the red orbit is substantially smaller than even the innermost blue orbit, and all 15 red trajectories converge to the same single small cycle regardless of their initial displacement. This result requires careful interpretation. Two reinforcing mechanisms are at work. First, the higher $\bar{u}_1 = 0.773$ relative to $\bar{u} = 0.750$ slightly reduces the local sensitivity of $\dot{z}$ to wage-share fluctuations: the slope of the $\dot{z} = 0$ locus at the equilibrium, proportional to $\gamma[1 - \tanh^2(-\gamma\bar{u})]$, falls from approximately 0.376 to 0.345

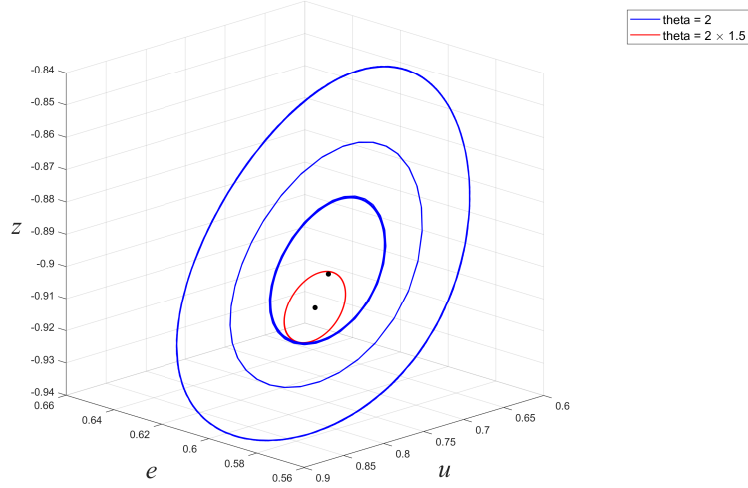


Figure 14: Effect of increasing the productivity damage parameter θ on the limit cycles in (e, u, z) -space. Blue orbits: baseline $\theta = 2$, with 15 trajectories initialized by displacing $(\bar{e}, \bar{u}, \bar{z})$ by ± 0.01 , ± 0.03 , -0.04 , and mixed perturbations along individual state-variable dimensions. Red orbit: $\theta = 3$, with 15 trajectories initialized by displacing the corresponding equilibrium $(\bar{e}_1, \bar{u}_1, \bar{z}_1)$ by the same perturbation patterns. Black dots indicate the respective equilibria P_1 for each parameterization. Other parameters: $v = 6.25$, $\alpha = 0.05$, $\beta_0 = 0.96$, $\beta_1 = 1.\bar{6}$, and $\gamma = 1.96$.

as $\bar{u}$ rises, meaning that cyclical swings in u translate into somewhat smaller movements in z . Second, and more importantly, the higher θ amplifies the productivity response to any given z deviation through the damage function $D(z) = \theta(1 + z)$, strengthening a self-correcting feedback loop that runs *from* the OUD index *back to* itself via employment and the wage share. When z rises above $\bar{z}_1$, a higher θ causes a larger fall in effective productivity growth $[1 - D(z)]\alpha$. Since output growth is pinned down by capital accumulation, slower productivity growth requires faster employment growth, so $\dot{e}/e$ rises more sharply. The tighter labor market accelerates real wage growth through the Phillips curve, raising the wage share u more forcefully; and since $\dot{z} = \tanh(-\gamma u) - z$, a higher u makes $\tanh(-\gamma u)$ more negative, pulling z back toward $\bar{z}_1$. The complete loop is $z \uparrow \rightarrow [1 - D(z)]\alpha \downarrow \rightarrow \dot{e}/e \uparrow \rightarrow u \uparrow \rightarrow z \downarrow$, a *negative* feedback that self-corrects any positive deviation of z from equilibrium. Higher θ amplifies the first link of this loop, and thereby strengthens the entire corrective chain. Crucially, the damage channel and the recovery channel are one and the same: more damaging OUD triggers faster employment and wage-share recovery, which reduces OUD more quickly. Together, the slightly weaker $u \rightarrow z$ channel and the stronger $z \rightarrow (e, u)$ corrective loop compress the cycle amplitude and eliminate the basin structure that supported multiple coexisting limit cycles under $\theta = 2$. A secondary consequence of this mechanism is that the coexistence of multiple limit cycles — visible for $\theta = 2$ as several concentric blue orbits — collapses under $\theta = 3$: all trajectories, regardless of their initial distance from the equilibrium, converge to a single small orbit. The high-damage regime therefore not only compresses cycle amplitude but also eliminates the history dependence documented in Section 4.2.

This result carries a sharp and non-trivial policy implication. Consider two economies that differ only in θ : one in which OUD has severe consequences for output per worker (high θ), and one in which the productivity impact is milder (low θ). The high- θ economy will exhibit a more drug-clean workforce at equilibrium and smaller cyclical swings in employment

and wages, but it will suffer larger productivity losses per OUD worker. The low- θ economy tolerates a higher equilibrium share of OUD workers and wider macroeconomic fluctuations, but with lower damage per affected individual. Interventions that reduce θ — such as workplace treatment programs, harm-reduction policies, or occupational redesign that lowers the productivity impact of OUD — may therefore inadvertently *widen* the growth cycle while improving individual health outcomes. A comprehensive policy assessment must weigh these macroeconomic dynamic effects alongside the direct welfare benefits of reducing the severity of substance abuse.

The contrasting effects of γ and θ on the orbit structure reflect their distinct positions in Figure 7. A reduction in γ weakens the *entry point* of the health channel: the arrow connecting income distribution to substance abuse. Because a smaller wage-share movement triggers a smaller OUD response, both the productivity effect and the employment effect in the health arm are attenuated uniformly across the cycle, shrinking the orbit while leaving the equilibrium composition only mildly affected. An increase in θ , by contrast, strengthens an *interior link* of the health channel: the arrow connecting OUD prevalence to productivity growth. Because this link is simultaneously the damage arrow (a rise in z slows productivity) and the recovery arrow (the same rise in z initiates the $z \rightarrow u \rightarrow z$ corrective loop), amplifying it compresses rather than widens the orbit. In both cases, the effective contribution of the health channel to the amplitude of oscillations is diminished, but through qualitatively different mechanisms: γ governs the width of the gate between economic stress and health behavior, while θ governs the strength of the productivity transmission once health behavior has changed — specifically, the sensitivity of employment growth to any given level of OUD prevalence, captured by $\partial(\dot{e}/e)/\partial z = \theta\alpha$.

5 Final considerations

This paper has extended Goodwin (1967)’s predator-prey growth-cycle model to incorporate the endogenous link between income distribution and substance abuse, and the feedback from health behavior to aggregate labor productivity. The extended framework introduces a population composition index z that evolves according to a continuous-time discrete-choice mechanism: as the wage share declines, the probability of substance use disorder rises among workers, and as the wage share recovers, health behavior improves. This index enters the economy through a productivity damage function $D(z)$, transforming what is typically treated as a static health externality into a dynamic macroeconomic channel that interacts with the investment and wage dynamics of the growth cycle.

The theoretical analysis establishes three main results. First, under mild regularity conditions, the three-dimensional nonlinear system admits a unique economically meaningful interior equilibrium. Second, applying the existence part of the Hopf bifurcation theorem, we show that as the autonomous wage-growth parameter β_0 increases through a critical threshold β_0^{HB} , the equilibrium loses local asymptotic stability and the system bifurcates into a family of periodic orbits. Third, numerical simulations above the bifurcation threshold reveal the coexistence of multiple stable limit cycles, so that the long-run amplitude of the growth cycle depends not only on structural parameters but also on the size and direction of initial macroeconomic conditions. Economies starting from larger employment imbalances are drawn into high-amplitude cycles with wider swings in wages, employment, and workforce health composition.

The sensitivity analysis yields two contrasting policy lessons. Reducing γ — the intensity of choice parameter governing how strongly substance-use behavior responds to income fluctuations — compresses the amplitude of the limit cycles and reduces the macroeconomic

instability generated by the health channel. Policies that weaken the link between economic stress and OUD, such as the expansion of social safety nets, universal access to mental health and substance-use treatment, and housing security programs, therefore carry a macroeconomic stabilizing dividend beyond their direct health benefits. Reducing θ —the productivity damage per OUD worker—does not unambiguously stabilize the cycle. While it lowers the level of productivity loss at the equilibrium, it also weakens a self-correcting feedback loop that runs from OUD-induced productivity damage back through employment and wage dynamics to OUD recovery, and can widen cycle amplitude while eliminating the coexistence of multiple attractors. Harm-reduction policies or workplace accommodation programs that reduce the on-the-job productivity cost of OUD should therefore be evaluated with awareness of this macroeconomic dynamic effect.

The empirical analysis in Section 2 provides grounding for the model’s key mechanisms, while also highlighting limits of the available evidence. The static cross-bin relationship between labor share shock severity and age-adjusted drug mortality is consistent with the income-compression channel assumed in the model. The dynamic local projections further suggest that positive labor share gains—particularly at moderate severity—reduce drug-induced mortality over horizons of one to four years. The response to negative shocks is less precisely estimated at severe levels, where treated observations are scarce, and near-zero at mild levels. Whether this asymmetry reflects a genuine nonlinearity in the health-behavior response or an identification limitation remains an open question.

Several extensions of the present framework merit future investigation. First, the model abstracts from the financial sector, treating investment as entirely profit-funded and ignoring debt dynamics. Incorporating Minsky’s financial instability hypothesis—wherein firms borrow to finance investment and debt accumulation amplifies or dampens the growth cycle—would allow the model to address how financial fragility interacts with the health-behavior channel, and whether periods of high indebtedness magnify the macroeconomic consequences of labor income compression. Second, the workforce is held constant and labor force participation is treated as separable from the productivity mechanism. A richer treatment would allow OUD prevalence to directly reduce the effective supply of labor, not only its productivity, amplifying employment dynamics during downturns in the spirit of [Krueger \(2017\)](#) and [Greenwood et al. \(2022\)](#). Third, the analysis is confined to an aggregate closed economy. A multi-region or open-economy extension could address how the geographic concentration of negative labor share shocks documented in the data interacts with the aggregate cycle. These extensions are left for future research.

A Mathematical Appendix

A.1 Proof of Proposition 1

Recall the equilibrium conditions:

$$\begin{aligned} 0 &= \left[\frac{1-u}{v} - (1-D(z))\alpha \right] e \\ 0 &= [f(e) - (1-D(z))\alpha] u \\ 0 &= \tanh(-\gamma u) - z \end{aligned}$$

An economically meaningful solution requires $e \neq 0$ and $u \neq 0$. Therefore, we have:

$$0 = \frac{1-u}{v} - (1-D(z))\alpha \quad (\text{A.1})$$

$$0 = f(e) - (1-D(z))\alpha \quad (\text{A.2})$$

$$z = \tanh(-\gamma u) \quad (\text{A.3})$$

From Eq. (A.1), we have that:

$$\begin{aligned} 1-u &= v\alpha - vD(\tanh(-\gamma u))\alpha \\ u - vD(\tanh(-\gamma u))\alpha &= 1 - v\alpha \end{aligned} \quad (\text{A.4})$$

Define the auxiliary function:

$$H(u) = u - vD(\tanh(-\gamma u))\alpha$$

Substituting in Eq. (A.4), we obtain:

$$H(u) = 1 - v\alpha$$

from which it follows that

$$\bar{u} = H^{-1}(1 - v\alpha)$$

provided that the inverse of $H(\cdot)$ exists.

Substituting $\bar{u}$ into Eq. (A.3), it follows that:

$$\bar{z} = \tanh(-\gamma H^{-1}(1 - v\alpha))$$

Moreover, substituting $\bar{u}$ and $\bar{z}$ into (A.2), we have:

$$\begin{aligned} f(e) &= \alpha[1 - D(\tanh(-\gamma u))] \\ \bar{e} &= f^{-1}(\alpha[1 - D(\tanh(-\gamma u))]) \\ &= f^{-1}(\alpha[1 - D(\tanh(-\gamma H^{-1}(1 - v\alpha))])) \end{aligned}$$

Still, the demonstration crucially depends on the existence of $H(\cdot)$ inverse. To prove this, we need to show whether $H(\cdot)$ is bijective. That is, injective, each input maps to a unique output, and surjective, every possible output is mapped from some input.

Proving injectivity: If a function f is differentiable on an interval I , and its derivative $f'(x)$ is always positive or always negative on that interval, then f is strictly monotonic, which implies injectivity. In our case:

$$\begin{aligned} H(u) &= u - \alpha v D(\tanh(-\gamma u)) \\ \frac{\partial H}{\partial u} &= 1 + \alpha v \gamma D'^2(-\gamma u) > 0 \end{aligned}$$

Thus, $H(u)$ is strictly increasing and injective.

Proving surjectivity: Using the Intermediate Value Theorem (IVT), if a function f is continuous on an interval $[a, b]$, and $f(a) < y < f(b)$ (or vice versa), then there exists some $c \in (a, b)$ such that:

$$f(c) = y$$

In our case, it requires to check limits at infinity:

$$\begin{aligned}\lim_{u \rightarrow 0} H(u) &= \lim_{u \rightarrow 0} u - v\alpha D(\tanh(-\gamma u)) = -vD(0)\alpha \\ \lim_{u \rightarrow 1} H(u) &= \lim_{u \rightarrow 1} u - v\alpha D(\tanh(-\gamma u)) = 1 - v\alpha D(\tanh(-\gamma))\end{aligned}$$

Thus, by the IVT, for every $y \in [-vD(0), 1 - v\alpha D(\tanh(-\gamma))]$, there exists $u \in [0, 1]$ such that $H(u) = y$. Therefore, $H(u)$ is surjective onto the interval $[-vD(0), 1 - v\alpha D(\tanh(-\gamma))]$.

A.2 Proof of Proposition 2

The Jacobian matrix is given by:

$$J = \begin{bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{bmatrix}$$

where

$$\begin{aligned}j_{11} &= \frac{\partial \dot{e}}{\partial e} \Big|_{\bar{e}, \bar{u}, \bar{z}} = 0 & j_{12} &= \frac{\partial \dot{e}}{\partial u} \Big|_{\bar{e}, \bar{u}, \bar{z}} = -\frac{\bar{e}}{v} & j_{13} &= \frac{\partial \dot{e}}{\partial z} \Big|_{\bar{e}, \bar{u}, \bar{z}} = \alpha \bar{e} D' \\ j_{21} &= \frac{\partial \dot{u}}{\partial e} \Big|_{\bar{e}, \bar{u}, \bar{z}} = \bar{u} f' & j_{22} &= \frac{\partial \dot{u}}{\partial u} \Big|_{\bar{e}, \bar{u}, \bar{z}} = 0 & j_{23} &= \frac{\partial \dot{u}}{\partial z} \Big|_{\bar{e}, \bar{u}, \bar{z}} = \alpha \bar{u} D' \\ j_{31} &= \frac{\partial \dot{z}}{\partial e} \Big|_{\bar{e}, \bar{u}, \bar{z}} = 0 & j_{32} &= \frac{\partial \dot{z}}{\partial u} \Big|_{\bar{e}, \bar{u}, \bar{z}} = -\gamma [1 - \tanh^2(-\gamma \bar{u})] & j_{33} &= \frac{\partial \dot{z}}{\partial z} \Big|_{\bar{e}, \bar{u}, \bar{z}} = -1\end{aligned}$$

Thus, the characteristic equation can be written as

$$\lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3 = 0$$

where the coefficients are given by

$$\begin{aligned}\text{(i)} \quad a_1 &= -\text{tr} J = -(0 + 0 - 1) = 1 \\ \text{(ii)} \quad a_2 &= \begin{vmatrix} 0 & \alpha \bar{u} D' \\ -\gamma [1 - \tanh^2(-\gamma \bar{u})] & -1 \end{vmatrix} + \begin{vmatrix} 0 & -\frac{\bar{e}}{v} \\ \bar{u} f' & 0 \end{vmatrix} = \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} \\ \text{(iii)} \quad a_3 &= -\det J = \bar{u} f' \left\{ \frac{\bar{e}}{v} + \alpha \bar{e} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] \right\} \\ &= \alpha \bar{e} \bar{u} \gamma D' f' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} \\ \text{(iv)} \quad a_1 a_2 - a_3 &= \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} - \alpha \bar{e} \bar{u} \gamma D' f' [1 - \tanh^2(-\gamma \bar{u})] - \bar{u} f' \frac{\bar{e}}{v} \\ &= \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] (1 - \bar{e} f')\end{aligned}$$

The necessary and sufficient conditions for the local stability of a given equilibrium point is that all roots of the characteristic equation have negative real parts. From the Routh-Hurwitz conditions, this requires:

$$\begin{aligned} a_1 &> 0 \\ a_2 &> 0 \\ a_3 &> 0 \\ a_1 a_2 - a_3 &> 0 \end{aligned}$$

where either the first or the second condition can be eliminated. For simplicity, we choose to evaluate conditions a_1 , a_2 , and $a_1 a_2 - a_3 > 0$. The following is the evaluation of the coefficients:

$$\begin{aligned} \text{(i)} \quad a_1 &= -\text{tr} J = -(0 + 0 + 1) = 1 > 0 \\ \text{(ii)} \quad a_2 &= \begin{vmatrix} 0 & \alpha \bar{u} D' \\ -\gamma([1 - \tanh^2(-\gamma \bar{u})] & -1 \end{vmatrix} + \begin{vmatrix} 0 & -\frac{\bar{e}}{v} \\ \bar{u} f' & 0 \end{vmatrix} = \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} > 0 \\ \text{(iii)} \quad a_3 &= -\det J = \bar{u} f' \left\{ \frac{\bar{e}}{v} + \alpha \bar{e} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] \right\} \\ &= \alpha \bar{e} \bar{u} \gamma D' f' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} > 0 \\ \text{(iv)} \quad a_1 a_2 - a_3 &= \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} - \alpha \bar{e} \bar{u} \gamma D' f' [1 - \tanh^2(-\gamma \bar{u})] - \bar{u} f' \frac{\bar{e}}{v} \\ &= \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] (1 - \bar{e} f') \geq 0 \end{aligned}$$

Therefore, the internal equilibrium P_1 is locally stable provided that:

$$1 - \bar{e} f' > 0$$

On the other hand, as long as a_1 , a_2 , and $a_3 > 0$, a Hopf bifurcation can occur for values of the parameters such that $a_1 a_2 - a_3 = 0$. That is when:

$$\bar{e} f' = 1$$

A.3 Proof of Proposition 3

Disregarding the trivial case in which $\bar{u} = \bar{z} = \bar{e} = 0$, the equilibrium of the dynamic system (14) imply:

$$\begin{aligned} 0 &= \frac{1-u}{v} - (1-D(z))\alpha \\ 0 &= f(e) - (1-D(z))\alpha \\ z &= \tanh(-\gamma u) \end{aligned}$$

Using the functional forms in (16), we can rewrite them as:

$$(1 - \theta(1+z))\alpha = \frac{1-u}{v} \tag{A.5}$$

$$(1 - \theta(1+z))\alpha = -\beta_0 + \beta_1 e \tag{A.6}$$

$$z = \tanh(-\gamma u) \tag{A.7}$$

By plugging (A.7) into (A.5), we have that:

$$\begin{aligned} 1 - u &= \alpha v(1 - \theta(1 + \tanh(-\gamma u))) \\ u - \alpha v \theta \tanh(-\gamma u) &= 1 - \alpha v + \alpha v \theta \end{aligned} \quad (\text{A.8})$$

Define the auxiliary function:

$$J(u) = u - \alpha v \theta \tanh(-\gamma u)$$

Substituting in Eq. (A.8), we obtain:

$$J(u) = 1 - \alpha v + \alpha v \theta$$

from which it follows that

$$\bar{u} = J^{-1}(1 - \alpha v + \alpha v \theta)$$

provided that the inverse of $J(\cdot)$ exists.

Substituting $\bar{u}$ into Eq. (A.7), it follows that:

$$\bar{z} = \tanh(-\gamma J^{-1}(1 - \alpha v + \alpha v \theta))$$

Moreover, substituting $\bar{z}$ into (A.6), we have:

$$\begin{aligned} \beta_1 \bar{e} &= \beta_0 + (1 - \theta(1 + \bar{z}))\alpha = \beta_0 + \alpha - \alpha\theta - \alpha\theta\bar{z} \\ \bar{e} &= \frac{\beta_0 + \alpha - \alpha\theta - \alpha\theta \tanh(-\gamma J^{-1}(1 - \alpha v + \alpha v \theta))}{\beta_1} \end{aligned}$$

Still, the demonstration crucially depends on the existence of $J(\cdot)$ inverse. To prove this, we need to show whether $J(\cdot)$ is bijective. That is, injective, each input maps to a unique output, and surjective, every possible output is mapped from some input.

Proving injectivity: If a function f is differentiable on an interval I , and its derivative $f'(x)$ is always positive or always negative on that interval, then f is strictly monotonic, which implies injectivity. In our case:

$$\begin{aligned} J(u) &= u - \alpha v \theta \tanh(-\gamma u) \\ \frac{\partial J}{\partial u} &= 1 + \alpha v \gamma \theta (1 - \tanh^2(-\gamma u)) > 0 \end{aligned}$$

Thus, $J(u)$ is strictly increasing given parameters $\alpha, v, \gamma, \theta$ are positive and injective.

Proving surjectivity: Using the Intermediate Value Theorem (IVT), if a function f is continuous on an interval $[a, b]$, and $f(a) < y < f(b)$ (or vice versa), then there exists some $c \in (a, b)$ such that:

$$f(c) = y$$

In our case, it requires to check limits at infinity:

$$\begin{aligned} \lim_{u \rightarrow 0} J(u) &= \lim_{u \rightarrow 0} u - \alpha v \theta \tanh(-\gamma u) = 0 \\ \lim_{u \rightarrow 1} J(u) &= \lim_{u \rightarrow 1} u - \alpha v \theta \tanh(-\gamma u) = 1 - \alpha v \theta \tanh(-\gamma) \end{aligned}$$

Thus, by the IVT, for every $y \in [0, 1 - \alpha v \theta \tanh(-\gamma)]$, there exists $u \in [0, 1]$ such that $J(u) = y$. Therefore, $J(u)$ is surjective onto the interval $[0, 1 - \alpha v \theta \tanh(-\gamma)]$.

A.4 Proof of Proposition 4

To prove Proposition 4 using the (existence part of) the Hopf Bifurcation Theorem and using β_0 as the bifurcation parameter, we must first (HB1) show that the characteristic equation possesses a pair of complex conjugate eigenvalues $\sigma(\beta_0) \pm i\omega(\beta_0)$ that become purely imaginary at the critical value $\beta_{0_{HB}}$ of the parameter — that is, $\sigma(\beta_{0_{HB}}) = 0$ — and that no other eigenvalues with zero real part exist at $\beta_{0_{HB}}$. Then, (HB2) we must check that the derivative of the real part of the complex eigenvalues with respect to the bifurcation parameter is nonzero at the critical value.

(HB1) Given that the conditions $a_1 > 0$, $a_2 > 0$ and $a_3 > 0$ are all satisfied, in order that the characteristic equation has one negative real root and a pair of complex roots with zero real part we must have:

$$a_1 a_2 - a_3 = 0$$

a condition which, given the expression for $a_1 a_2 - a_3$, is satisfied for

$$\begin{aligned} f'(\bar{e}) * \bar{e} &= \beta_1 * \frac{\beta_0 + \alpha - \alpha\theta - \alpha\theta \tanh(-\gamma J^{-1}(1 - \alpha v + \alpha v\theta))}{\beta_1} \\ &= \beta_0 + \alpha - \alpha\theta - \alpha\theta \tanh(-\gamma J^{-1}(1 - \alpha v + \alpha v\theta)) = 1 \end{aligned}$$

(HB2) By using the so-called sensitivity analysis, it is then possible to show that the second requirement of the Hopf Bifurcation Theorem is also met. Substituting the elements of the Jacobian matrix into the respective coefficients of the characteristic equation:

$$\begin{aligned} a_1 &= 1 \\ a_2 &= \alpha \bar{u} \gamma D' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} = \alpha \bar{u} \gamma \theta [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} \beta_1 \frac{\bar{e}}{v} \\ a_3 &= \alpha \bar{e} \bar{u} \gamma D' f' [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} f' \frac{\bar{e}}{v} = \alpha \bar{e} \bar{u} \gamma \theta \beta_1 [1 - \tanh^2(-\gamma \bar{u})] + \bar{u} \beta_1 \frac{\bar{e}}{v} \end{aligned}$$

so that

$$\begin{aligned} \frac{\partial a_1}{\partial \beta_0} &= 0 \\ \frac{\partial a_2}{\partial \beta_0} &= \frac{\bar{u}}{v} > 0 \\ \frac{\partial a_3}{\partial \beta_0} &= \alpha \bar{u} \gamma \theta [1 - \tanh^2(-\gamma \bar{u})] + \frac{\bar{u}}{v} > 0 \end{aligned}$$

When $\bar{e} f'(\bar{e}) = 1$, apart from $a_1 > 0$, $a_2 > 0$, and $a_3 > 0$ which is always true, one also has $a_1 a_2 - a_3 = 0$. In this case, one root of the characteristic equation is real negative (λ_1), whereas the other two are a pair of complex roots with zero real part ($\lambda_{2,3} = \sigma \pm i\omega$, with $\sigma = 0$). We thus have:

$$\begin{aligned} a_1 &= -(\lambda_1 + \lambda_2 + \lambda_3) \\ &= -(\lambda_1 + 2\sigma) \end{aligned}$$

$$\begin{aligned} a_2 &= \lambda_1 \lambda_2 + \lambda_1 \lambda_3 + \lambda_2 \lambda_3 \\ &= 2\lambda_1 \sigma + \sigma^2 + \omega^2 \end{aligned}$$

$$\begin{aligned} a_3 &= -\lambda_1 \lambda_2 \lambda_3 \\ &= -\lambda_1 (\sigma^2 + \omega^2) \end{aligned}$$

such that:

$$\begin{aligned}
\frac{\partial a_1}{\partial \beta_0} &= -\frac{\partial \lambda_1}{\partial \beta_0} - 2\frac{\partial \sigma}{\partial \beta_0} = 0 \\
\frac{\partial a_2}{\partial \beta_0} &= 2\sigma\frac{\partial \lambda_1}{\partial \beta_0} + 2(\lambda_1 + \sigma)\frac{\partial \sigma}{\partial \beta_0} + 2\omega\frac{\partial \omega}{\partial \beta_0} = \frac{\bar{u}}{v} \\
\frac{\partial a_3}{\partial \beta_0} &= -(\sigma^2 + \omega^2)\frac{\partial \lambda_1}{\partial \beta_0} - 2\lambda_1\sigma\frac{\partial \sigma}{\partial \beta_0} - 2\lambda_1\omega\frac{\partial \omega}{\partial \beta_0} \\
&= \alpha\bar{u}\gamma\theta [1 - \tanh^2(-\gamma\bar{u})] + \frac{\bar{u}}{v}
\end{aligned}$$

For $\theta = 0$, the system to be solved becomes:

$$\begin{aligned}
-\frac{\partial \lambda_1}{\partial \beta_0} - 2\frac{\partial \sigma}{\partial \beta_0} &= 0 \\
2\lambda_1\frac{\partial \sigma}{\partial \beta_0} + 2\omega\frac{\partial \omega}{\partial \beta_0} &= \frac{\bar{u}}{v} \\
-\omega^2\frac{\partial \lambda_1}{\partial \beta_0} - 2\lambda_1\omega\frac{\partial \omega}{\partial \beta_0} &= \alpha\bar{u}\gamma\theta [1 - \tanh^2(-\gamma\bar{u})] + \frac{\bar{u}}{v}
\end{aligned}$$

or

$$\begin{bmatrix} -1 & -2 & 0 \\ 0 & 2\lambda_1 & 2\omega \\ -\omega^2 & 0 & -2\lambda_1\omega \end{bmatrix} \begin{bmatrix} \frac{\partial \lambda_1}{\partial \beta_0} \\ \frac{\partial \sigma}{\partial \beta_0} \\ \frac{\partial \omega}{\partial \beta_0} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{\bar{u}}{v} \\ \alpha\bar{u}\gamma\theta [1 - \tanh^2(-\gamma\bar{u})] + \frac{\bar{u}}{v} \end{bmatrix}$$

Thus:

$$\begin{aligned}
\left. \frac{\partial \sigma}{\partial \beta_0} \right|_{\beta_0 = \beta_{0\text{HB}}} &= \frac{\begin{vmatrix} -1 & 0 & 0 \\ 0 & \frac{\bar{u}}{v} & 2\omega \\ -\omega^2 & \alpha\bar{u}\gamma\theta [1 - \tanh^2(-\gamma\bar{u})] + \frac{\bar{u}}{v} & -2\lambda_1\omega \end{vmatrix}}{\begin{vmatrix} -1 & -2 & 0 \\ 0 & 2\lambda_1 & 2\omega \\ -\omega^2 & 0 & -2\lambda_1\omega \end{vmatrix}} \\
&= \frac{(-1)(-2\lambda_1\omega\frac{\bar{u}}{v} - 2\omega\alpha\bar{u}\gamma\theta[1 - \tanh^2(-\gamma\bar{u})] - 2\omega\frac{\bar{u}}{v})}{4\omega(\lambda_1^2 + \omega^2)} \\
&= \frac{2\omega(\lambda_1\frac{\bar{u}}{v} + \frac{\bar{u}}{v} + \alpha\bar{u}\gamma\theta[1 - \tanh^2(-\gamma\bar{u})])}{4\omega(\lambda_1^2 + \omega^2)} \\
&= \frac{2\omega((1 + \lambda_1)\frac{\bar{u}}{v} + \alpha\bar{u}\gamma\theta[1 - \tanh^2(-\gamma\bar{u})])}{4\omega(\lambda_1^2 + \omega^2)} > 0??
\end{aligned}$$

B Additional numerical experiments

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